Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 15 & 16

Clinical Anatomy of the Lower Limb 2 – Thigh, Knee Joint, Leg

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3D Lower Limb Model on Sketchfab

[Lower Limb Model](https://sketchfab.com/3d-models/lower-limb-by-compartment-822b2c5750c44b02a6f2851a8ea906ac)

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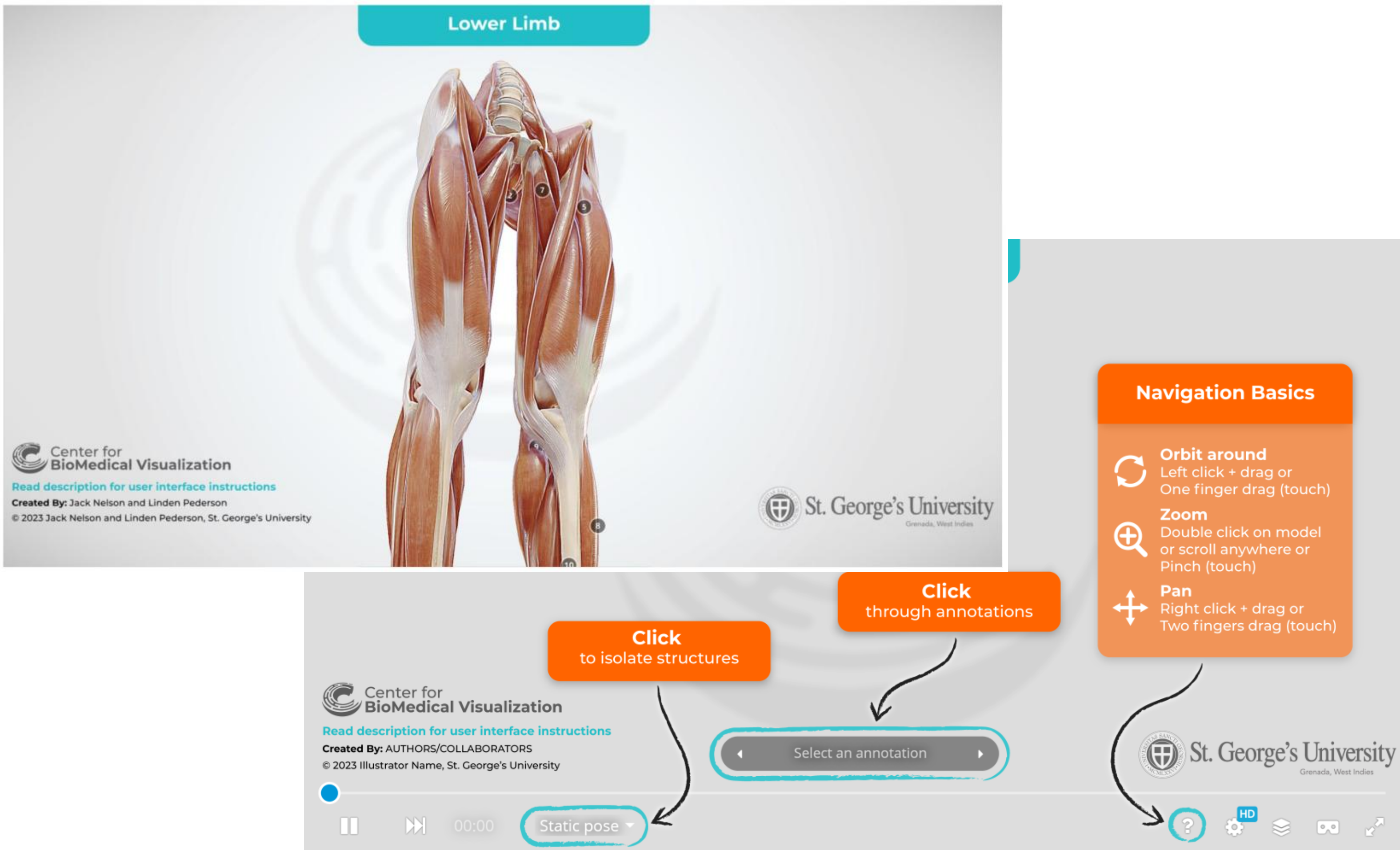

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. Newest ed. Elsevier Churchill Livingstone.

- Online Textbook: Chapter 6
  - Regional Anatomy:
    - Thigh
    - Knee joint
    - Popliteal fossa
    - Leg

- Pay special attention to the green **"In the Clinic"** boxes for the clinical correlations for this topic

Objectives

SOM.MK.II.BPM1.1.MSK.1.ANAT.1305	Describe the structures in the femoral sheath and their relationship to one another.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1306	List the boundaries and the contents of the femoral triangle, adductor canal and popliteal fossa.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1307	Describe the course of a femoral hernia and how it differs from an inguinal hernia.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1308	Discuss the consequences of piriformis syndrome, paying particular attention to the course of sciatic nerve through the piriformis muscle (SG Article)
SOM.MK.II.BPM1.1.MSK.1.ANAT.1310	Describe the movements that occur at the knee joint
SOM.MK.II.BPM1.1.MSK.1.ANAT.1311	Describe the knee joint its cartilage and ligamentous support
SOM.MK.II.BPM1.1.MSK.1.ANAT.1312	Describe Genu Valgus and Varus
SOM.MK.II.BPM1.1.MSK.1.ANAT.1313	Describe the function of the anterior and posterior cruciate ligaments.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1314	Describe the anterior and posterior drawer sign and the ligaments tested during each.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1315	Describe the function of the collateral ligaments and menisci of the knee joint
SOM.MK.II.BPM1.1.MSK.1.ANAT.1316	Describe the “unhappy triad” in knee joint injury.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1317	Describe the significance of the prepatellar, suprapatellar and infrapatellar bursae.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1318	Describe the blood supply of the knee joint.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1319	Describe the crural fascia.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1320	List the muscle compartments of the leg and describe their general attachments, actions and neurovascular supply.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1321	Explain the consequence of a lesion of the tibial nerve in the popliteal fossa.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1322	Explain the consequence of a lesion of the common fibular nerve at the neck of the fibula.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1323	Explain the consequence of a lesion of the superficial and deep fibular nerves just distal to their origin.

Overview

- Femoral sheath, femoral triangle, adductor canal
- Arterial blood supply of the thigh
- Venous drainage of thigh
- Knee joint
  - Cartilages and ligaments
  - Arterial blood supply
- Posterior leg
  - Posterior compartment muscles and their innervation
  - Popliteal fossa
- Anterior compartment of the leg
  - Anterior compartment muscles and their innervation
- Lateral compartment of the leg
  - Lateral compartment muscles and their innervation
- Cutaneous innervation of leg
- Arterial blood supply of leg
- Venous drainage of leg
- Nerve injuries

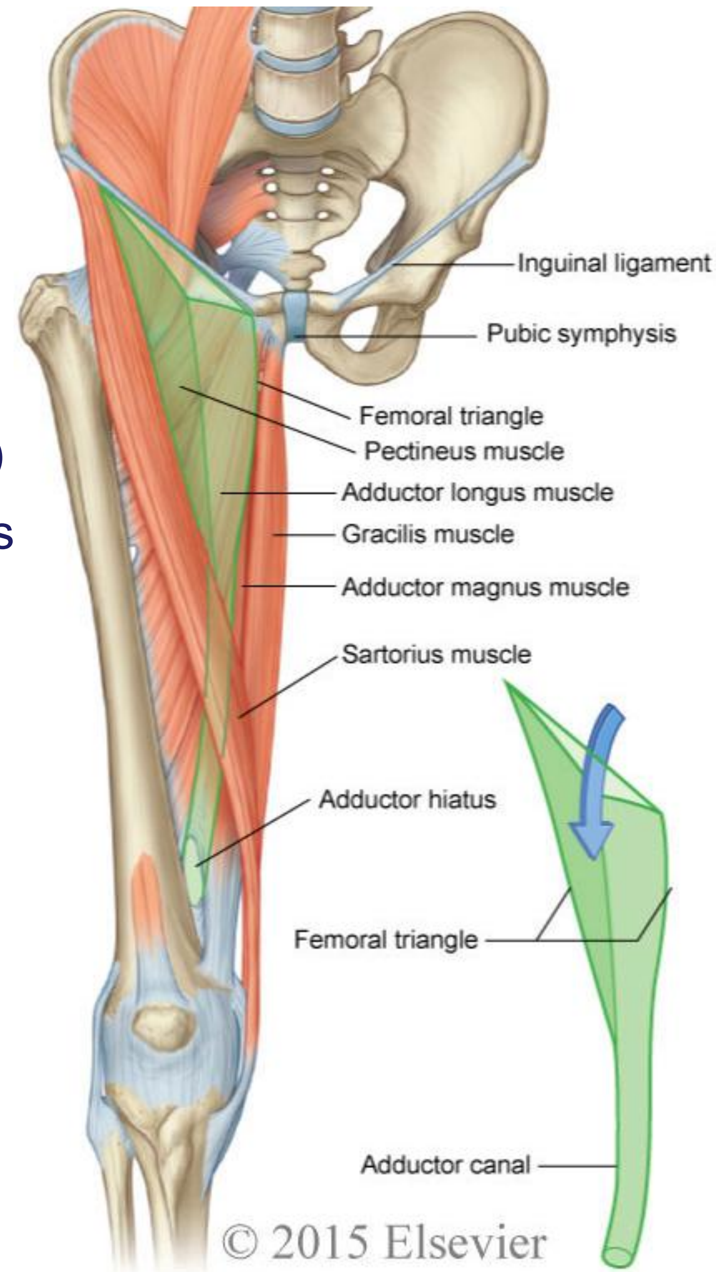
Femoral sheath, femoral triangle, adductor canal

Femoral Triangle

- A wedge-shaped depression formed by muscles in the upper thigh
- Contains structures coming from the abdomen

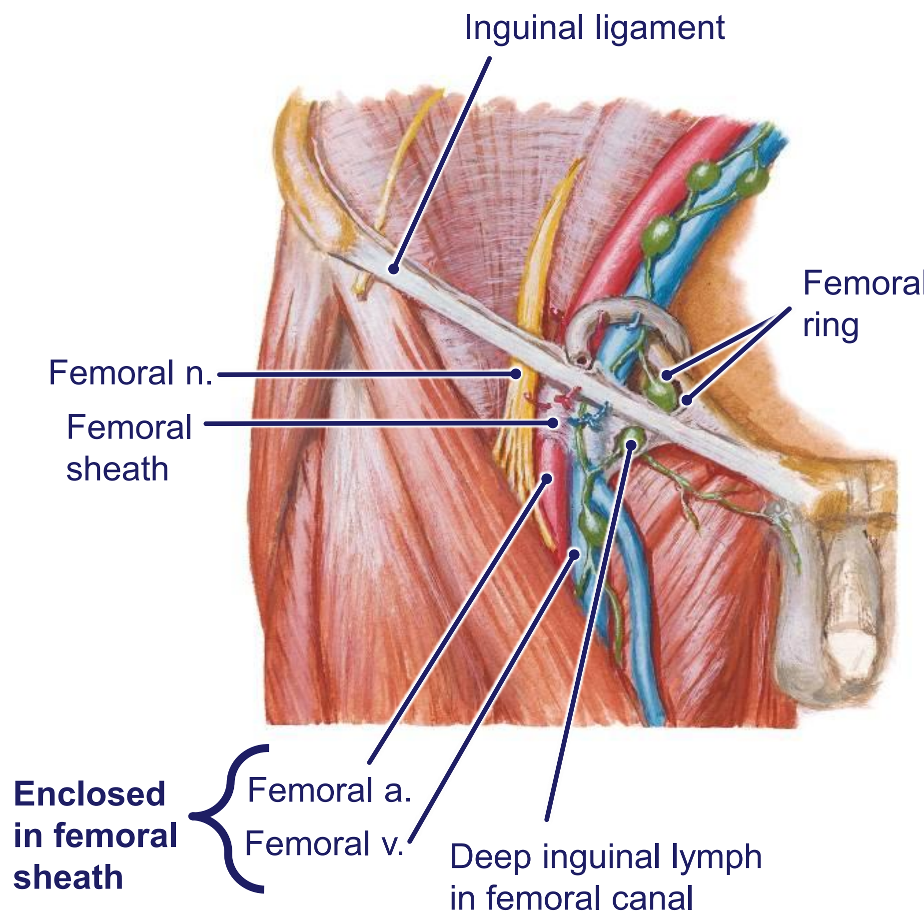
Boundaries:

- **Superior:** inguinal ligament (forms the base)
- **Medial:** medial border of the adductor longus
- **Lateral:** medial border of the sartorius
- **Roof:** fascia lata
- **Floor:** pectineus, iliopsoas and the adductor longus



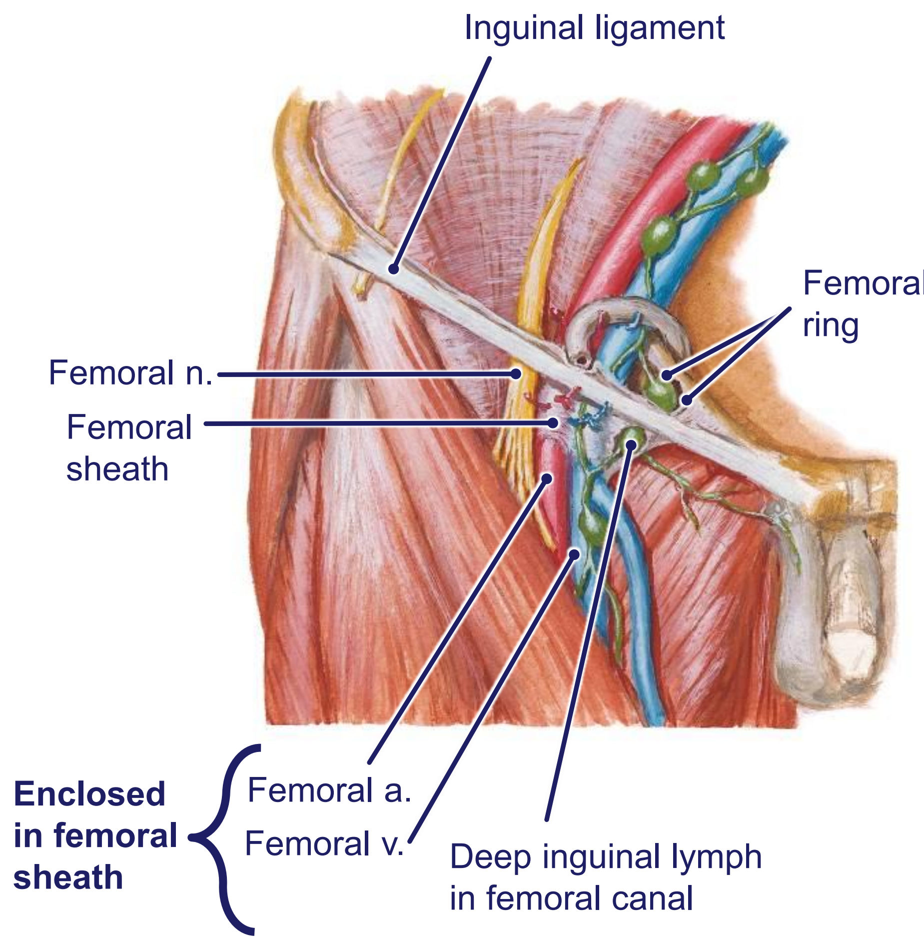
Femoral Triangle

- Contents (lateral to medial):
- Femoral nerve
- Femoral sheath
  - Femoral artery
  - Femoral vein
  - Femoral canal
    - Deep inguinal lymph nodes



Femoral Sheath

- Inferior extension of the transversalis fascia and the iliopsoas fascia from the abdomen.
- Divided internally into three smaller compartments by connective tissue:
    - The lateral compartment - the **femoral artery**
    - The intermediate compartment - the **femoral vein**
    - The most medial compartment - the **femoral canal** - contains **deep inguinal lymph nodes.**
        - **Femoral ring:** opening of the femoral canal superiorly, a potentially a weak point in the lower abdomen and is **the site for femoral hernias.**
    - The femoral nerve is lateral to but **not** contained within the femoral sheath.

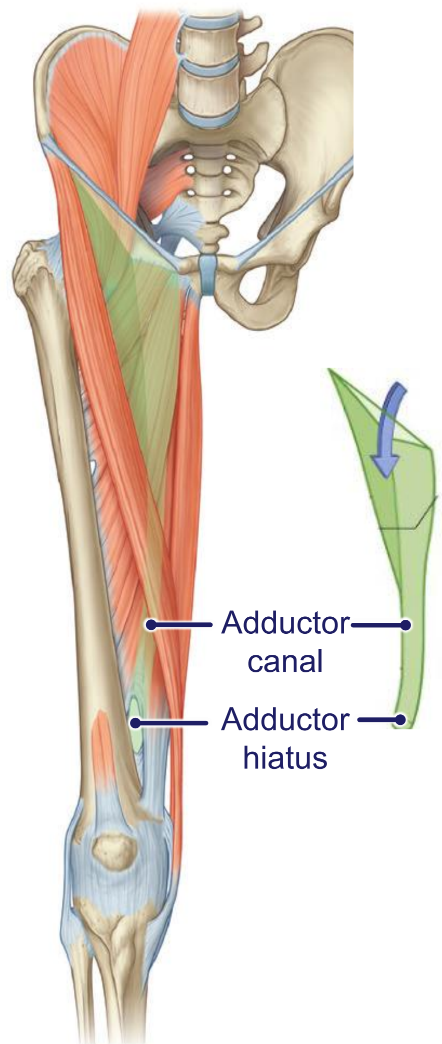


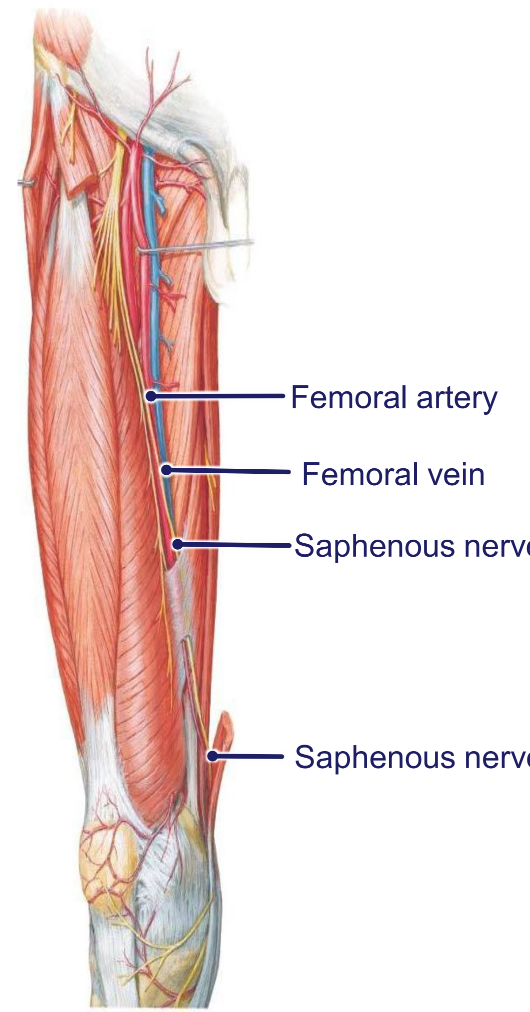
Adductor Canal

- **Boundaries:**
  - **Anteromedial** – Sartorius
  - **Lateral** – Vastus medialis
  - **Posterior** – Adductor longus and adductor magnus
- Fascial tunnel that continues from the apex of the femoral triangle
- Descends medially and posteriorly down the thigh to the **adductor hiatus**, **opening into the popliteal fossa.**
- **Femoral vessels exits** the adductor hiatus as the **popliteal vessels**
- Saphenous nerve although in the adductor canal, does not enter the adductor hiatus. It *leaves* the canal and continues its course to the medial side of the lower extremity

Contents:

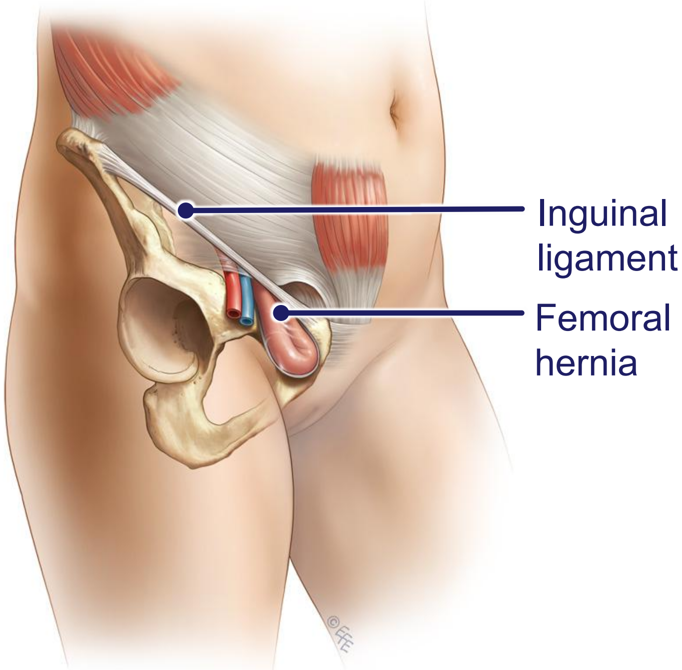
- Femoral artery
- Femoral vein
- Saphenous nerve





Femoral Hernia

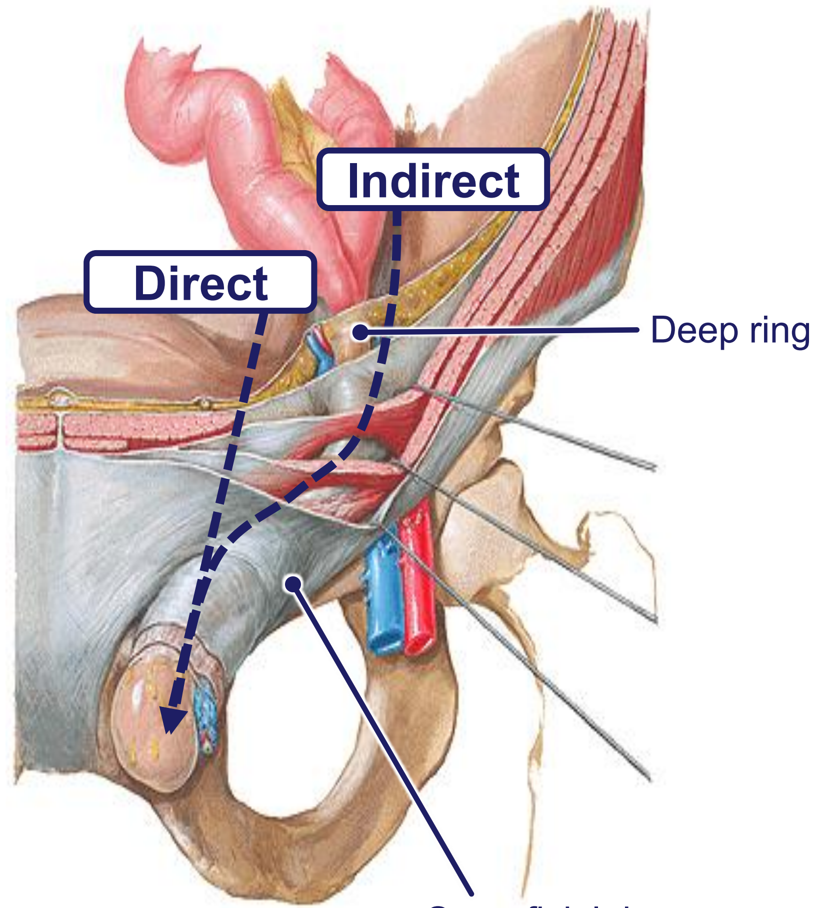
Originates **below** the inguinal ligament



- Usually occurs at the femoral ring
- Femoral sac protrudes into the femoral canal
- Palpated as a mass inferolateral to the pubic tubercle
- If large enough can pass through the saphenous opening into the subcutaneous tissue of the thigh

Inguinal Hernia

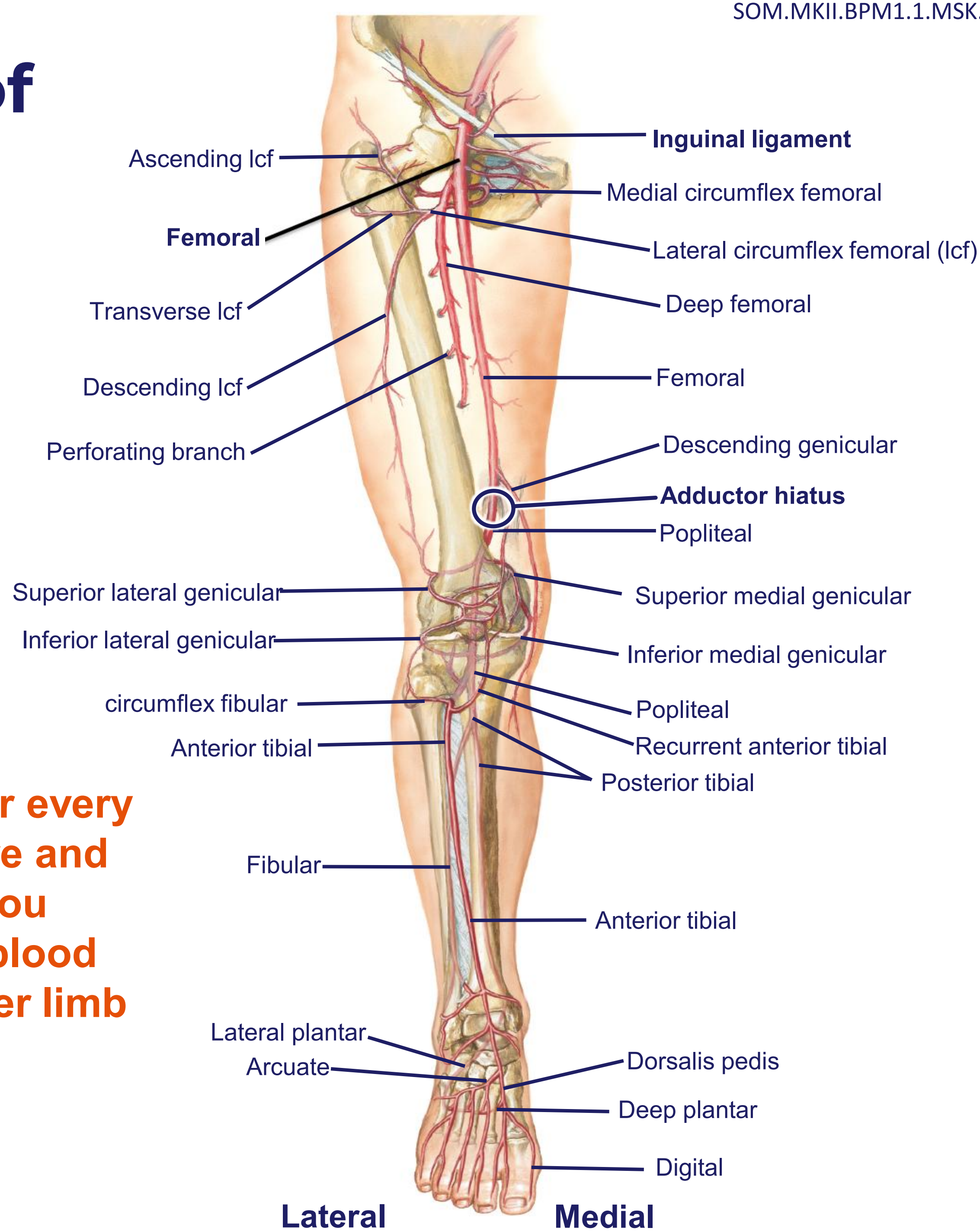
Originates **above** the inguinal ligament



Arterial blood supply of thigh

St. George's University
Grenada, West Indies

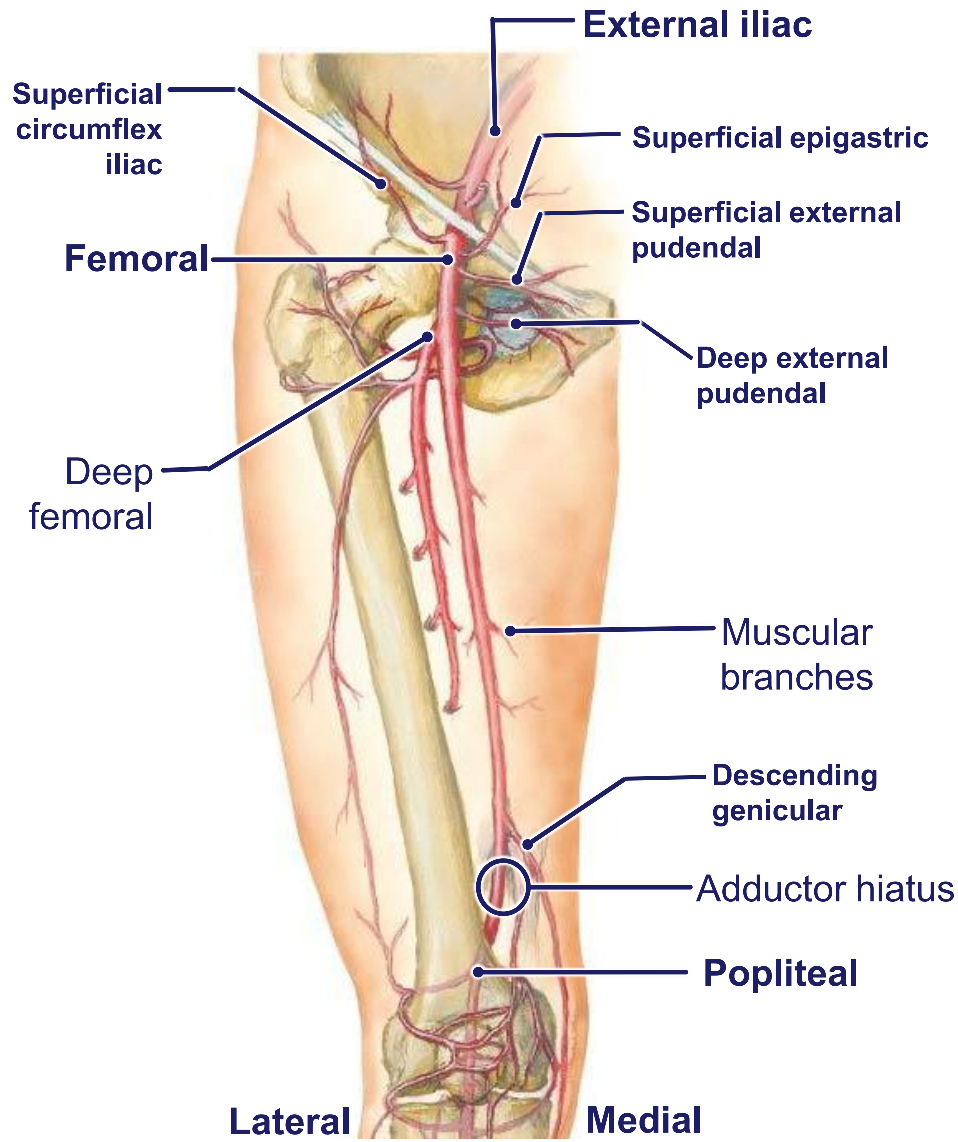
Summary of arterial supply of lower limb



Use this slide after every lower limb lecture and DLA to help you summarize the blood supply to the lower limb

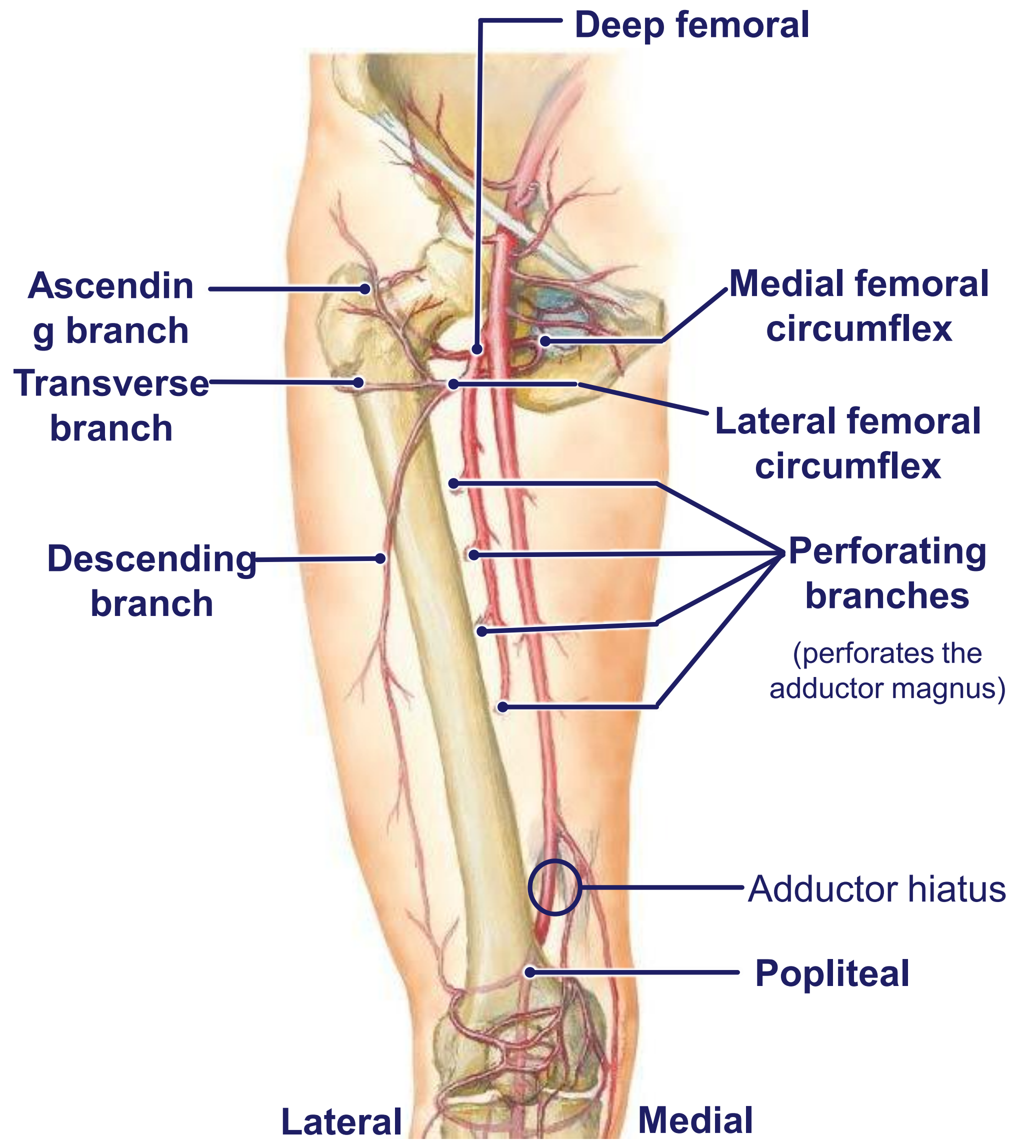
Blood supply to Thigh – Femoral Artery

- Continuation of external iliac as it passes beneath inguinal ligament into the femoral triangle
- Gives blood supply to thigh, hip, anterior abdominal wall and perineum through various branches
- Continues down the thigh in the adductor canal
- Exits the thigh into the popliteal fossa via the adductor hiatus, becoming the **popliteal**



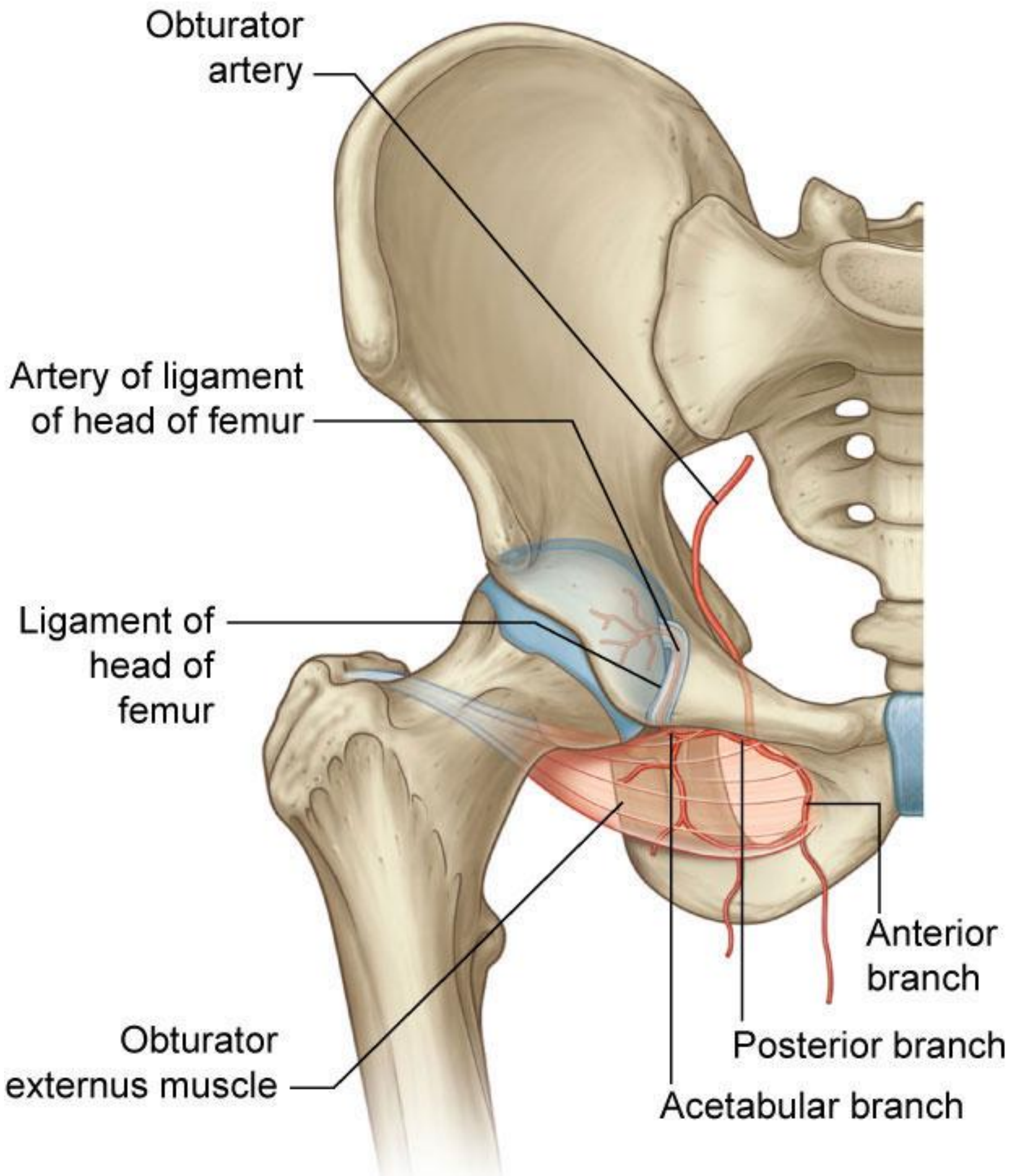
Blood supply to Thigh – Deep Femoral

- Largest branch of the femoral artery
- **Lateral circumflex femoral artery:**
  - Passes deep to sartorius and rectus femoris
  - Ascending branch
  - Descending branch
  - Transverse branch
- **Medial circumflex femoral artery:**
  - One branch ascends to the trochanteric fossa
  - One passes laterally to participate in the cruciate anastomosis
- **Perforating arteries:**
  - Penetrate adductor magnus and supply posterior compartment
  - Ascending and descending branches that interconnect forming a longitudinal channel

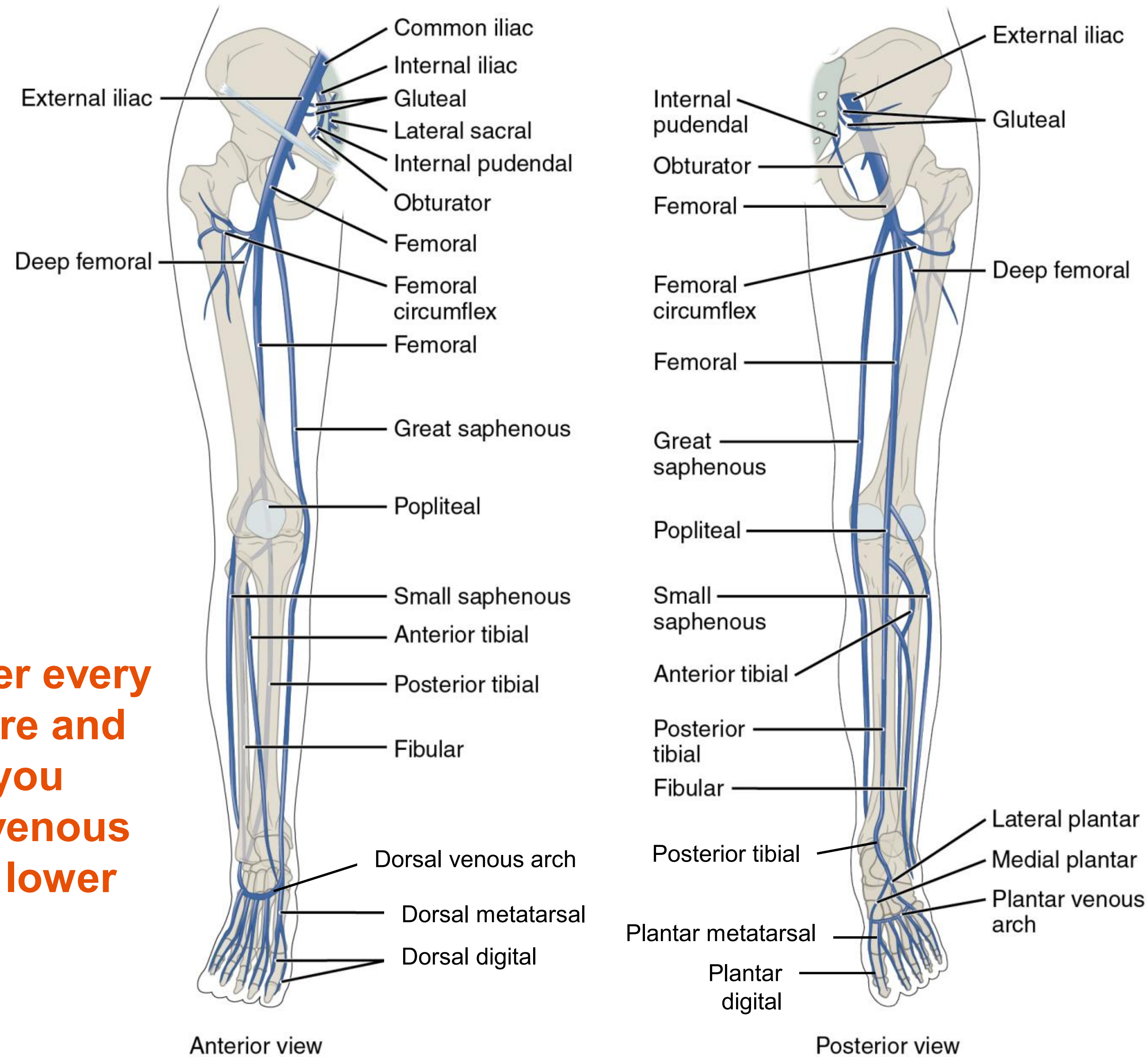


Blood supply to the Thigh – Obturator

- **Originates from the internal iliac**
- Enters the medial thigh via the obturator canal
- Anterior and posterior branches
    - Supplies the muscles of the medial compartment
- Contributes to blood supply to the head of the femur via the ligament of the head of the femur (from the acetabular branch)

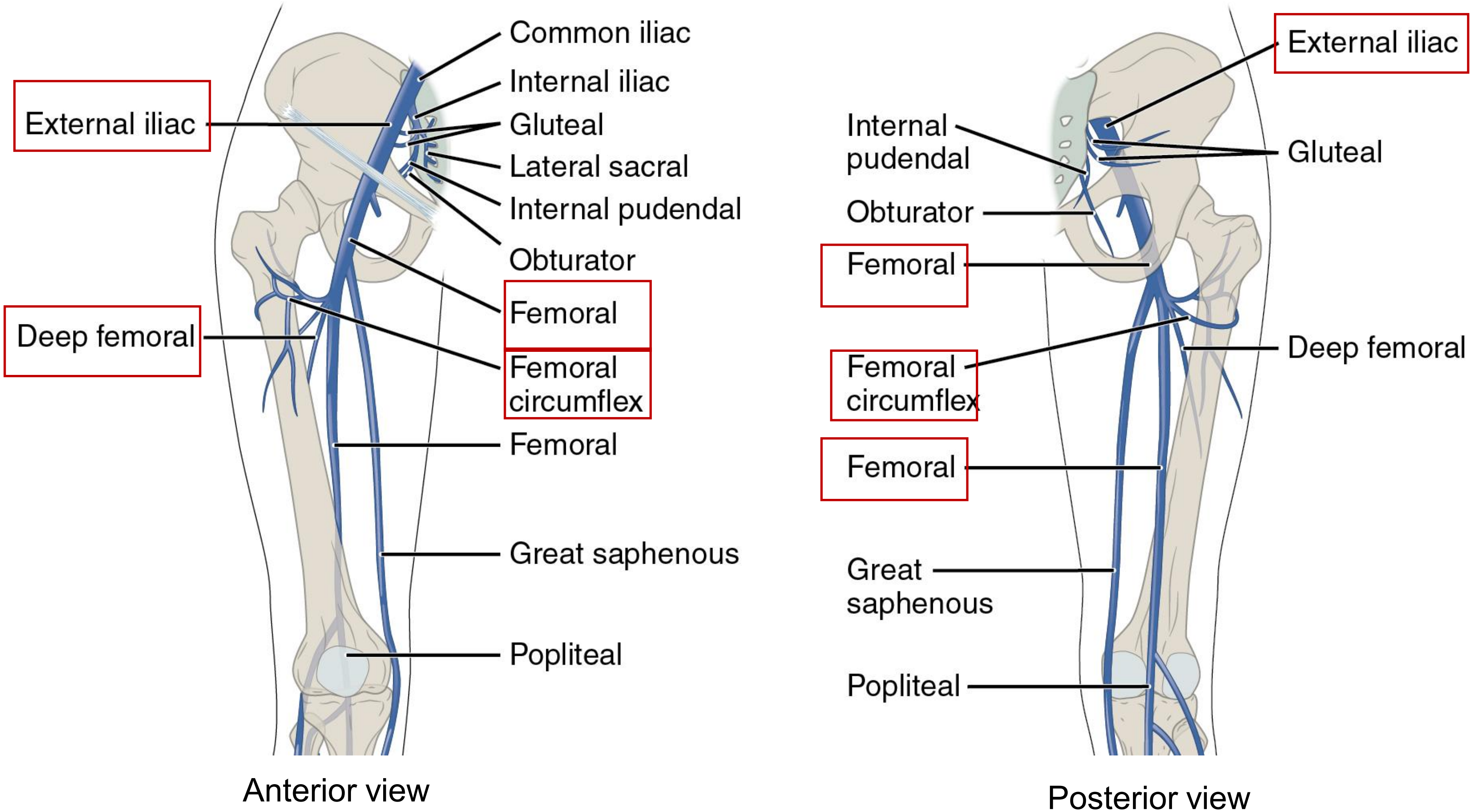


Summary of venous drainage of lower limb



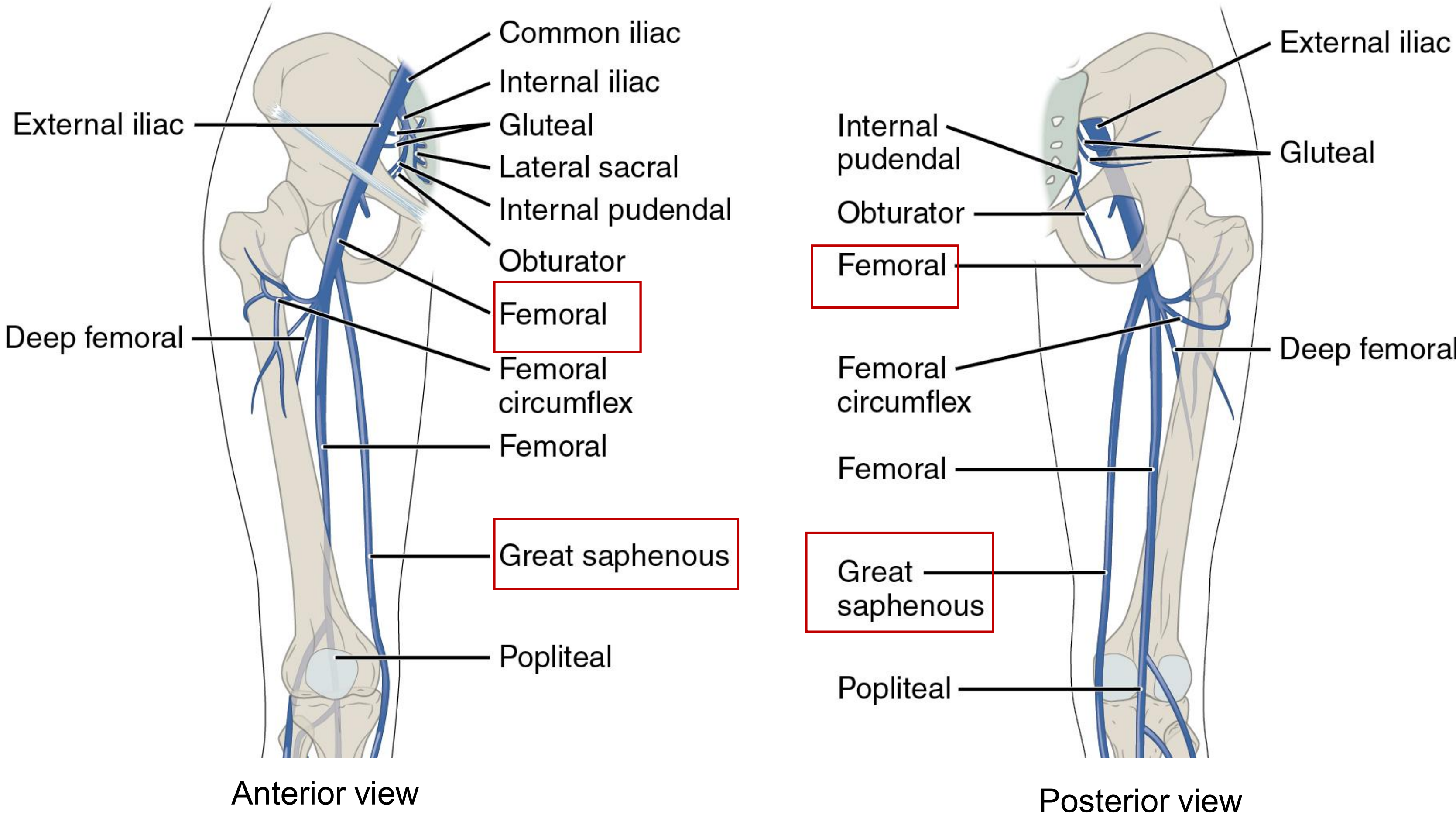
Use this slide after every lower limb lecture and DLA to help you summarize the venous drainage to the lower limb

Venous Drainage of Thigh – Deep Veins



- Located in the compartments of the thigh
- **Runs with the corresponding arteries - same name and usually paired**
- **Femoral circumflex** veins drain into **deep femoral** vein. **Deep femoral vein** drains into **femoral vein. Femoral vein** becomes the **external iliac vein** at the inguinal ligament.

Venous Drainage of Thigh – Superficial Veins

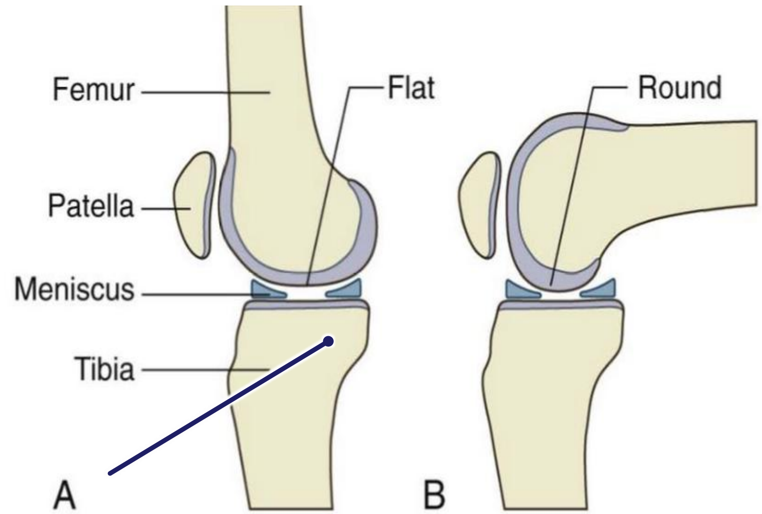


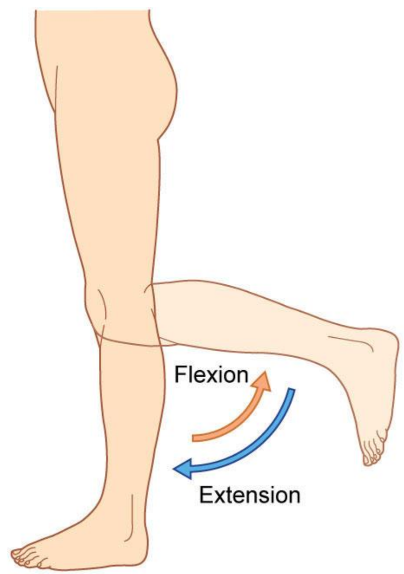
- Located in the skin and superficial fascia of the thigh.
- **Great saphenous vein** drains into the **femoral vein** at the saphenous opening in the fascia lata.

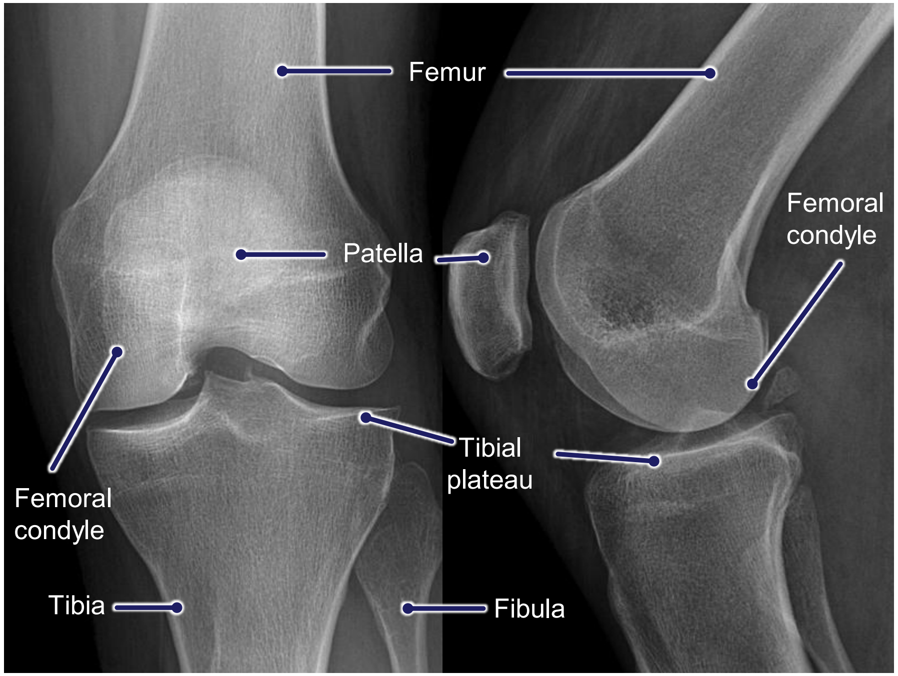
Knee joint

Knee Joint

- **Largest synovial joint in the body**
- Type: **Hinge**
  - articulation between **the femoral condyles** and **superior articular surfaces** (also called the plateau) of **tibial condyles**
  - articulation between the patella and the femur
- Allows flexion and extension







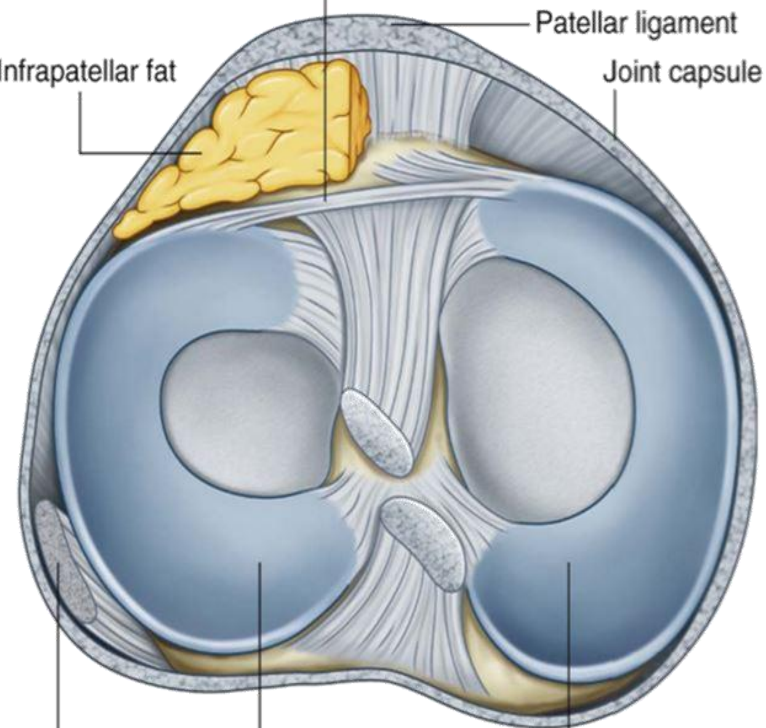
Anterior-Posterior View

Lateral View

Cartilages of the Knee - Menisci

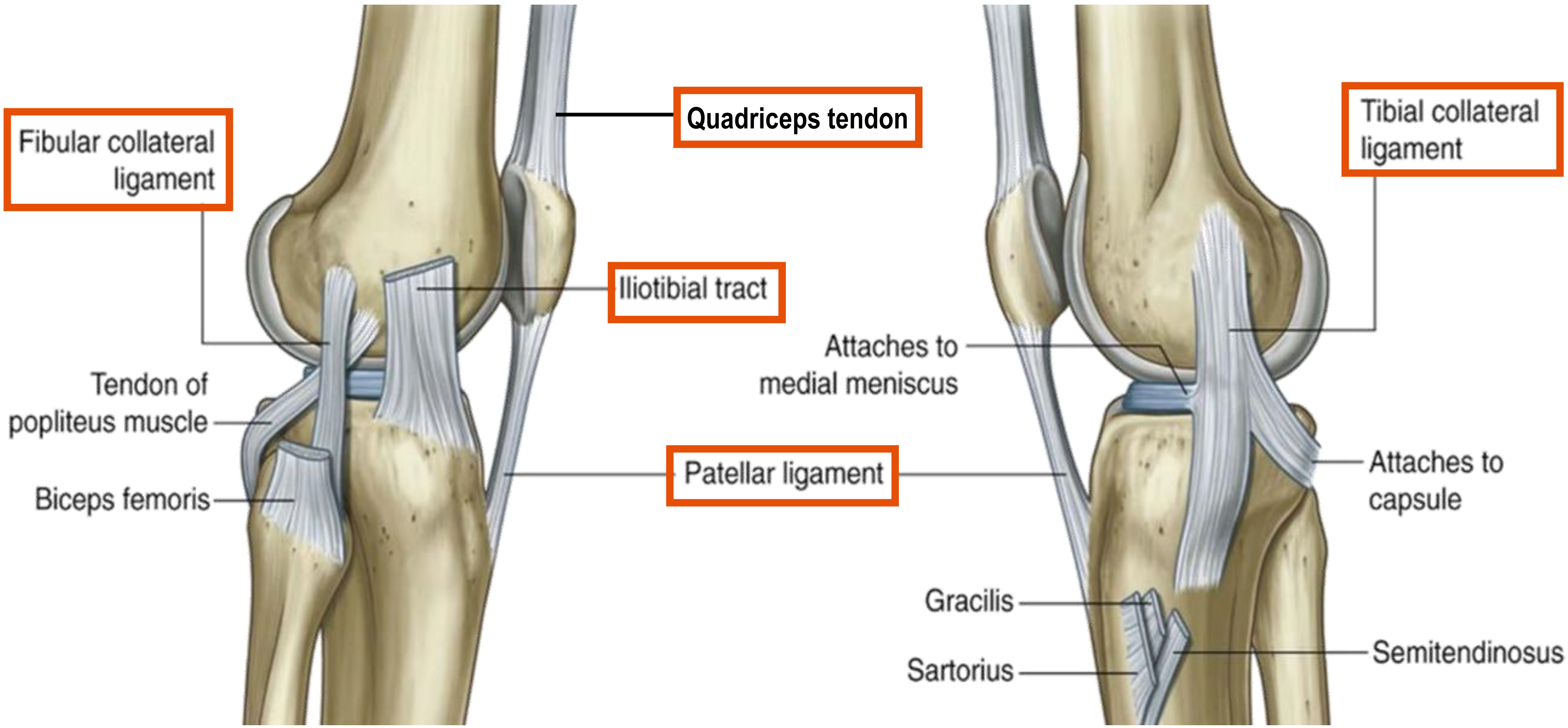
- **Two menisci: fibrocartilaginous C-shaped cartilages**
  - Menisci increase the congruency between the tibial and femoral condyles
  - Attached medially to the intercondylar region of the tibia
  - Medial meniscus: attached around its margin to the capsule of the joint and to the tibial collateral ligament
  - Lateral meniscus: unattached to the capsule as tendon of the popliteus muscle passes between them.

- The menisci are interconnected anteriorly by a transverse ligament of the knee



Collateral Ligaments of the Knee

Patellar ligament: the continuation of the quadriceps femoris tendon inferior to the patella



Fibular (lateral) collateral ligament

- Lateral femoral condyle to the fibular head
- Stabilizes the lateral side of the knee in varus stress

Tibial (medial) collateral ligament

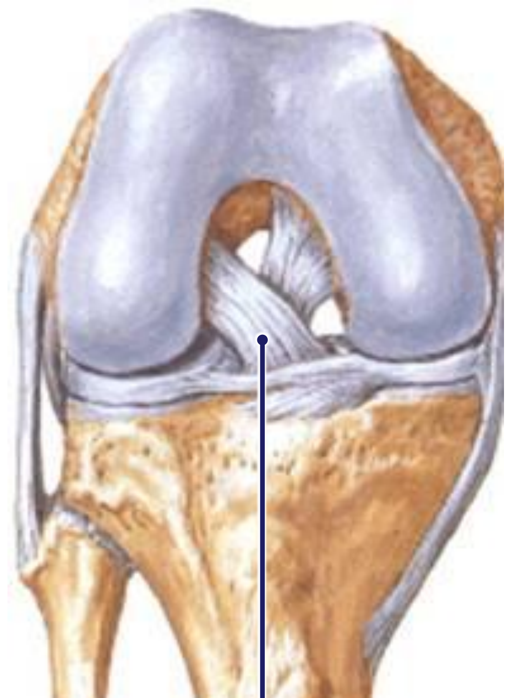
- Medial femoral condyle to the medial aspect of tibia (deep fibers are attached to the medial meniscus)
- Stabilizes the medial side of the knee in valgus stress

Anterior Cruciate Ligament

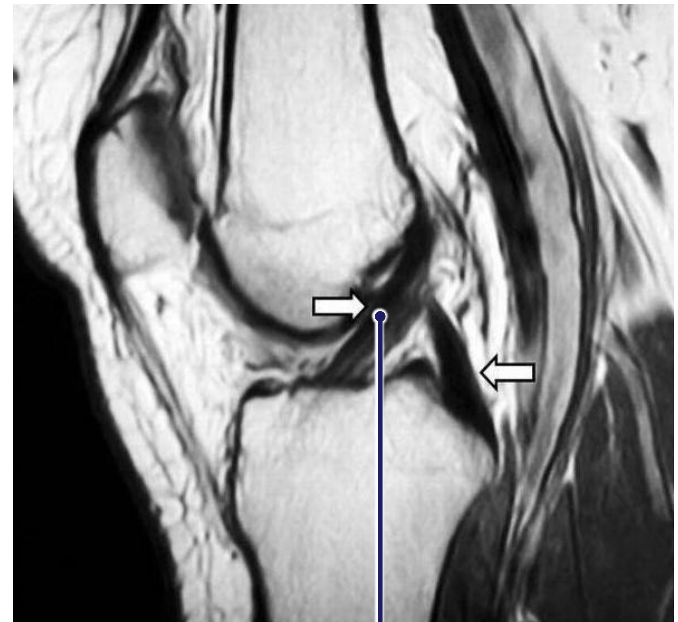
- Attaches to the anterior part of the intercondylar area of the tibia and ascends posteriorly to the lateral wall of the intercondylar fossa of the femur

- **Prevents anterior displacement of the tibia relative to the femur**
- **Prevents posterior displacement of the femur**
- **Anterior drawer test**

Anterior view



Lateral view



https://upload.wikimedia.org/wikipedia/commons/b/bd/MRT_ACL_PCL_01.jpg

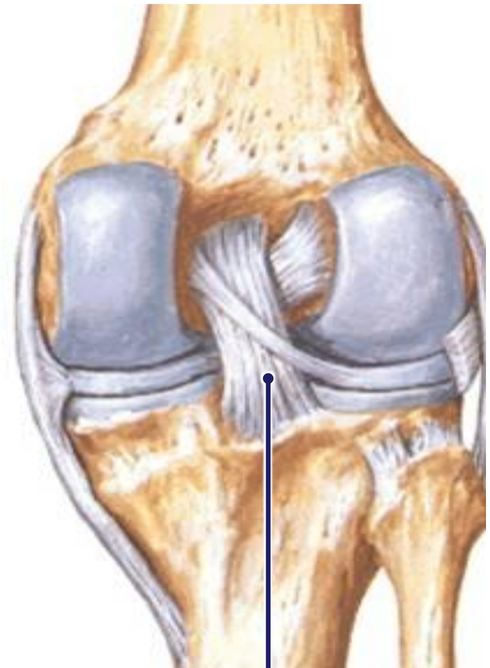
anterior cruciate

Posterior Cruciate Ligament

- Attaches to the posterior aspect of the intercondylar area of the tibia and ascends anteriorly to attach to the medial wall of the intercondylar fossa of the femur

- **Restricts posterior displacement of the tibia relative to the femur**
- **Restricts anterior displacement of the femur**
- **Posterior drawer test**

Posterior view



Lateral view

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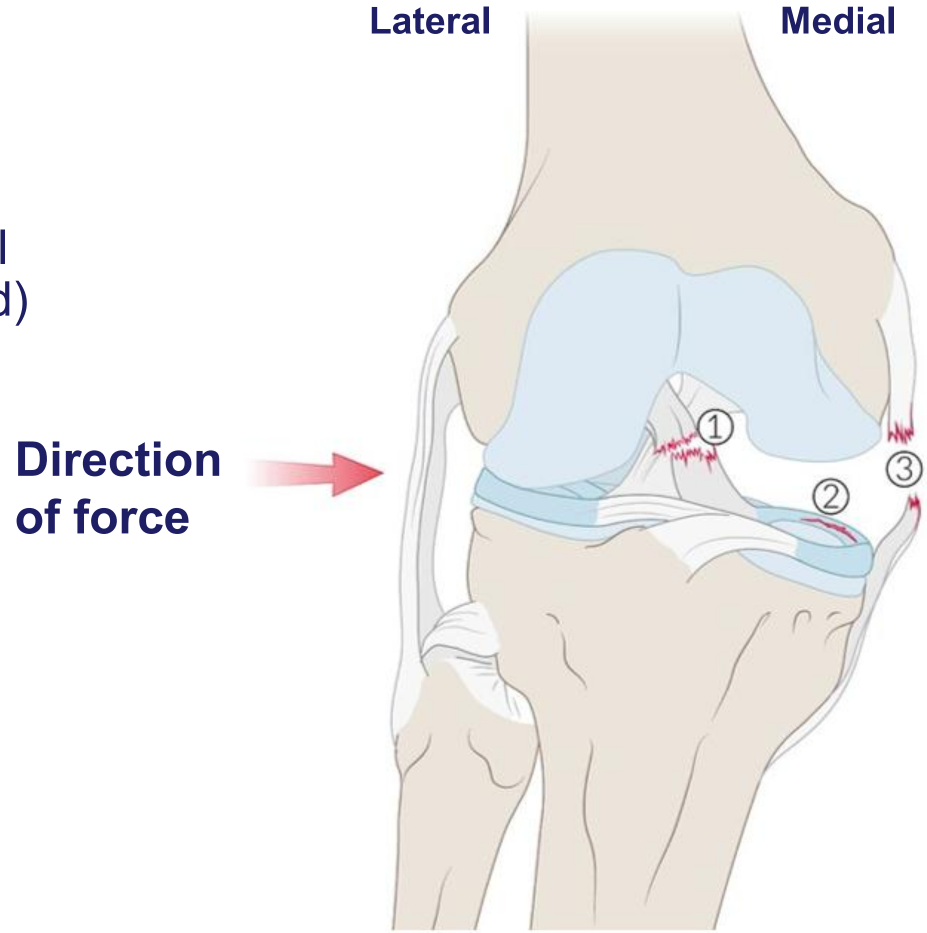
posterior cruciate

Unhappy Triad

Excessive valgus pressure to the knee:

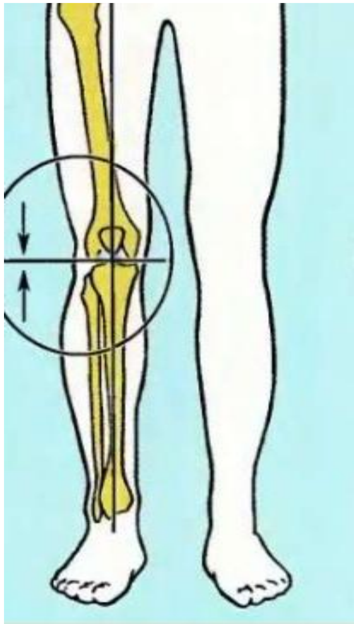
1. Anterior cruciate
2. Medial meniscus
3. Tibial (medial) collateral (most commonly injured)



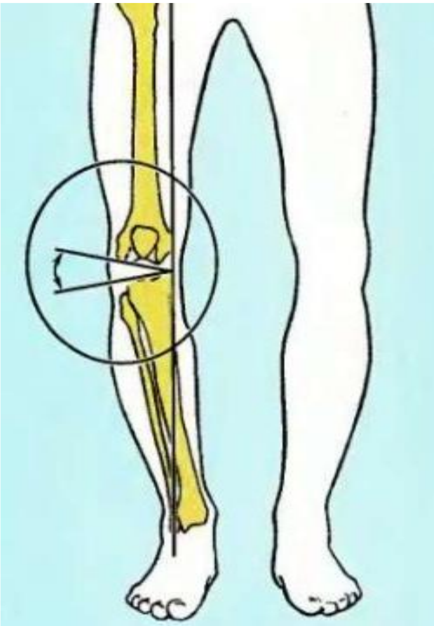
Anatomie O'Donoghue unhappy triad. Uit: https://www.amboss.com/us/knowledge/Knee_ligament_injuries/

Genu Vara and Genu Valga

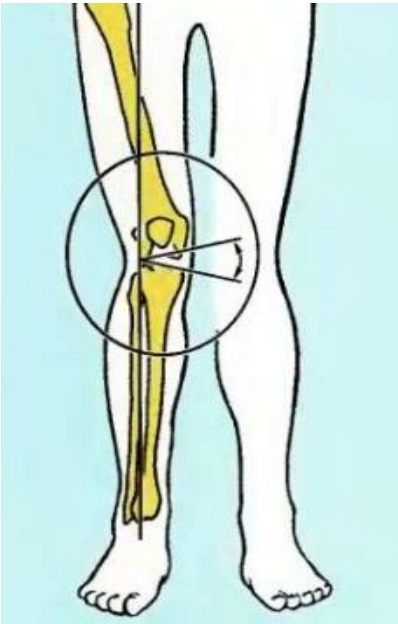
Normal



Genu vara: distal end of the tibia is directed inward (knees directed outward)



Genu valga: distal end of the tibia is directed outward (knees directed inward)

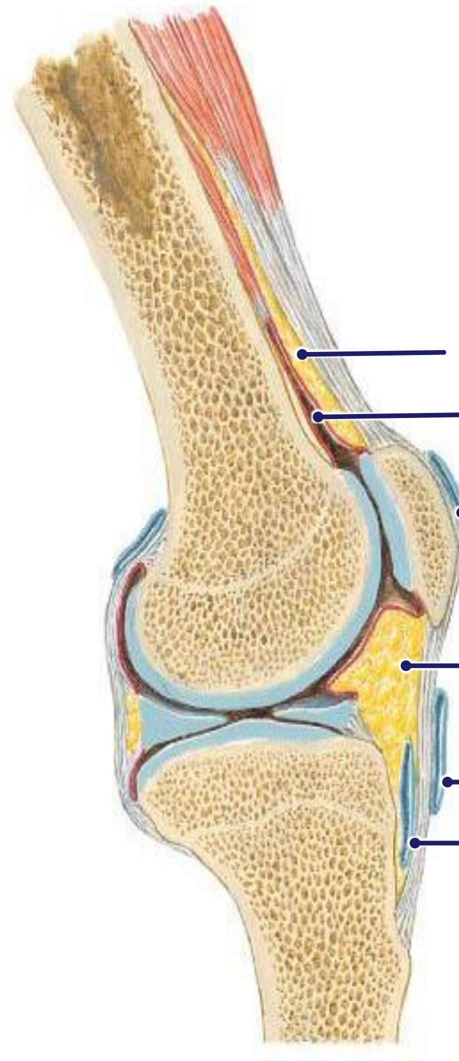


Knee Bursae

- Suprapatellar bursa: continuous with the articular cavity superiorly between the distal end of the shaft of the femur and the quadriceps femoris muscles and tendon.

- Other bursae associated with the knee but not normally communicating with the articular cavity include:
    - Subcutaneous prepatellar bursa
    - Subcutaneous infrapatellar bursa
    - Deep infrapatellar bursa

- Infrapatellar fat pad: a cylindrical layer of fat separating the synovial membrane from the patellar ligament.



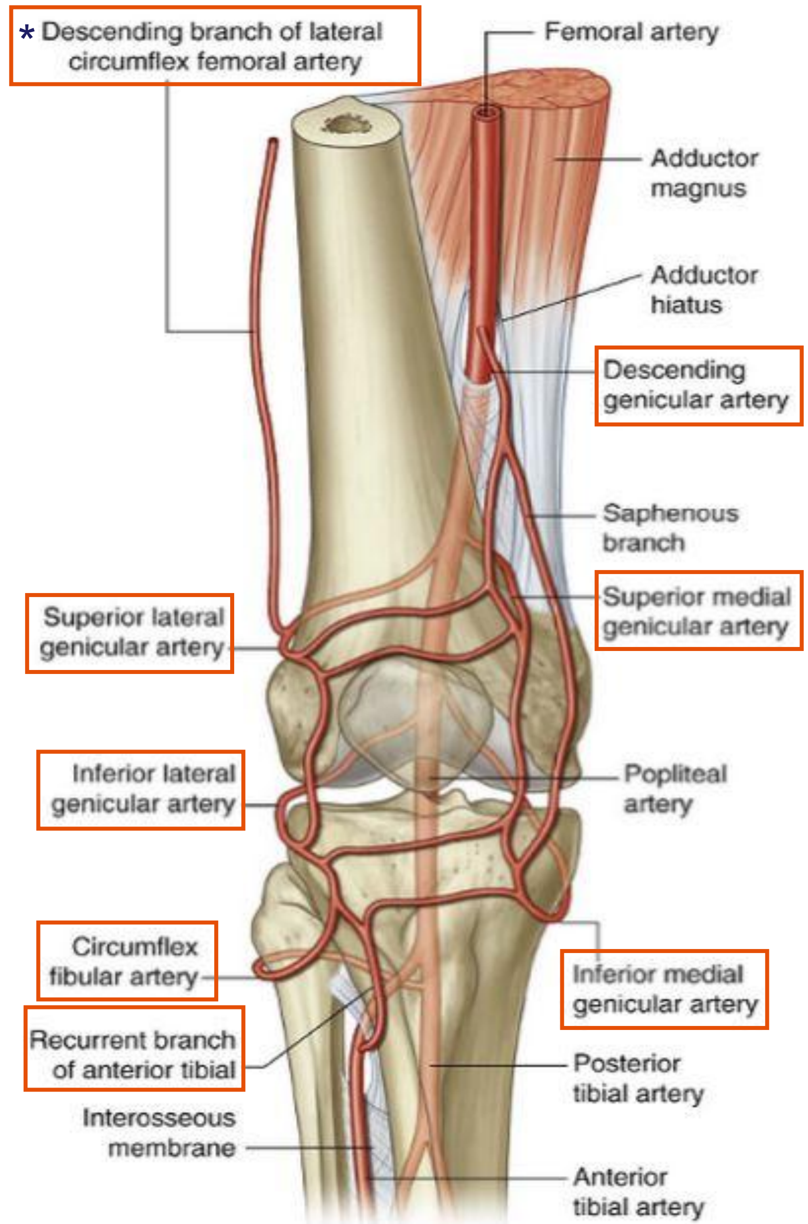
Prepatellar Bursitis

- **Inflammation** of the subcutaneous prepatellar bursa
- Commonly **caused by chronic irritation** due to trauma or repetitive kneeling. Can also be caused by infections such as gout.
- Patients complain of **swelling, pain and decreased range of motion.**
- **Calcified debris** in the subcutaneous prepatellar bursa in chronic inflammation.

Blood Supply to the Knee Joint

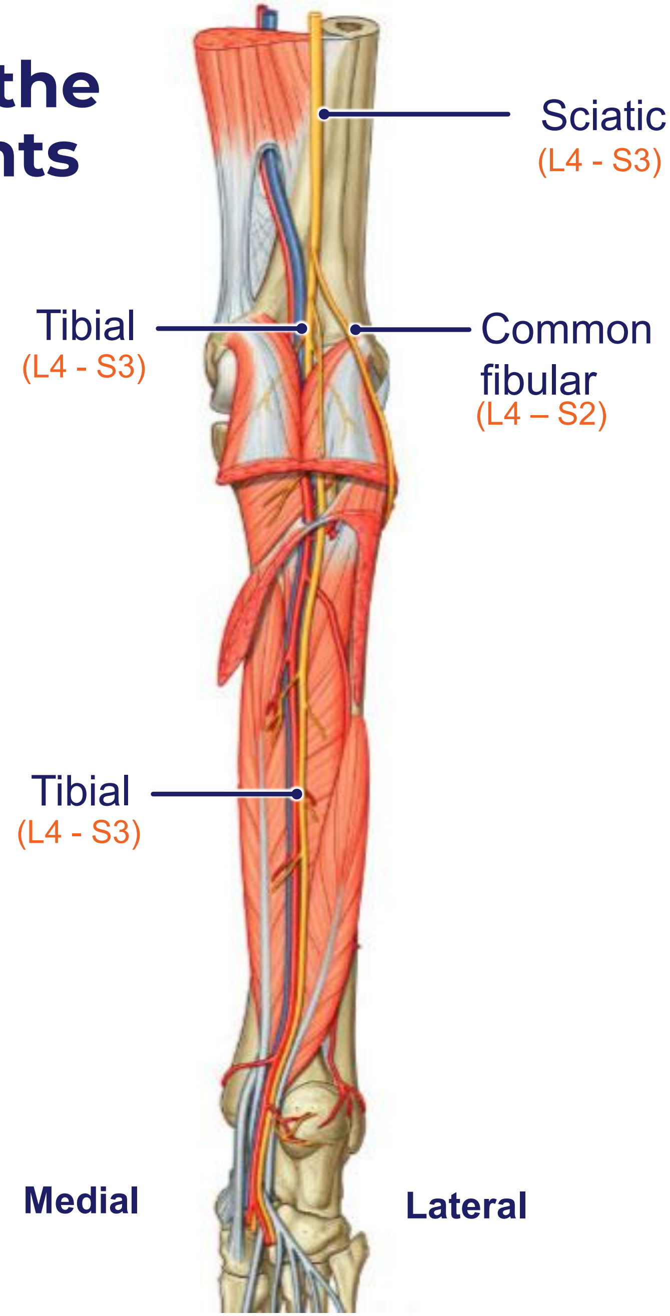
Anastomotic network of vessels around the joint provides collateral blood supply

- Genicular arteries from the popliteal artery
  - superior medial genicular
  - superior lateral genicular
  - inferior medial genicular
  - inferior lateral genicular
- Descending genicular from femoral artery
- Recurrent branches of tibial arteries
- Circumflex fibular
- *Descending branch of lateral circumflex femoral artery

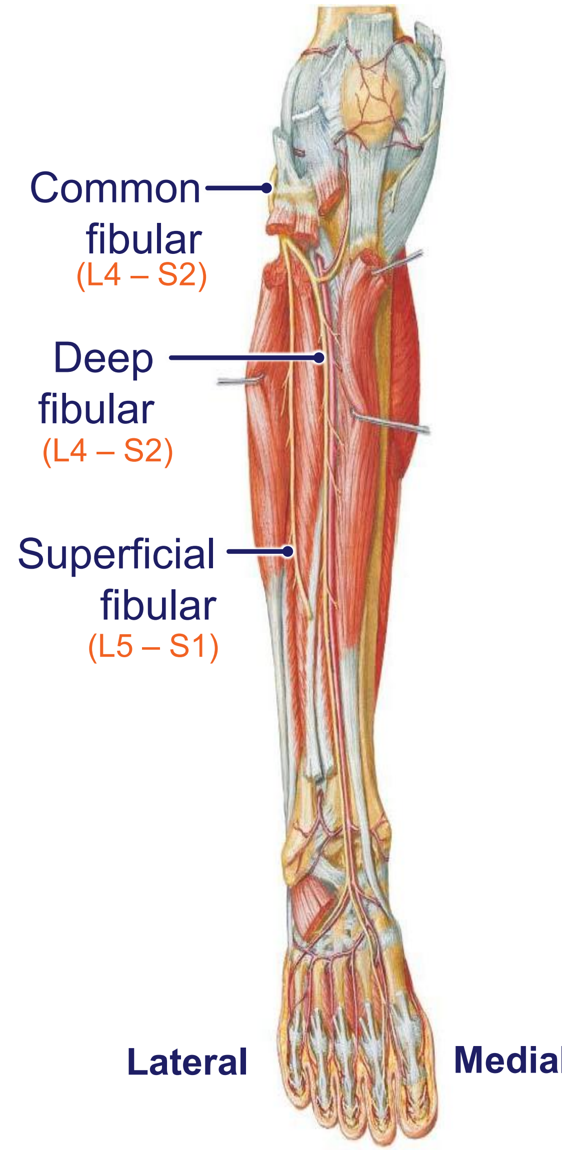


Compartments of the leg

Nerves innervating the compartments of the leg



Posterior view



Anterior view

Summary of Compartments of the Leg

Divided into three compartments by:

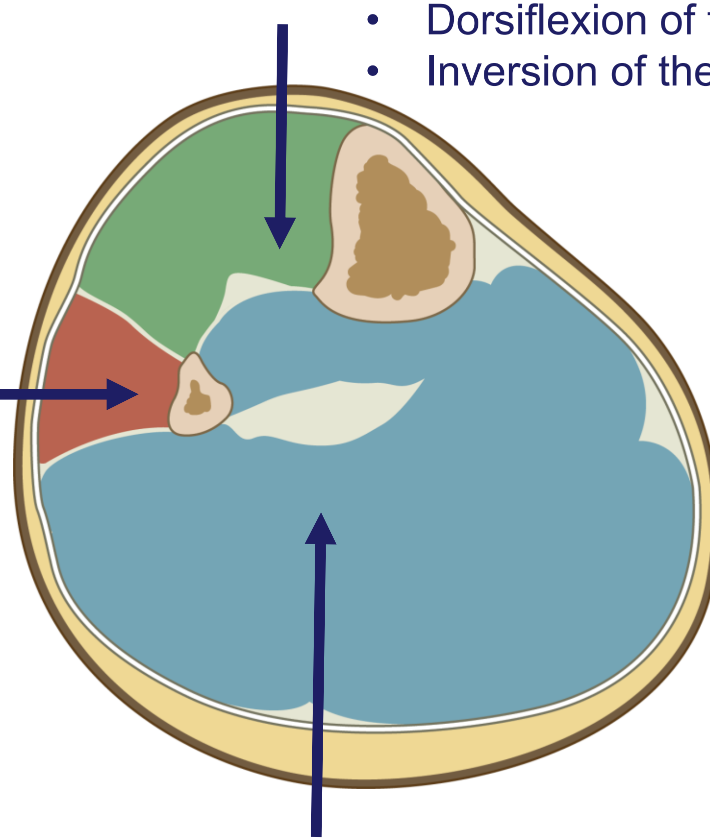
- Surrounding fascia crura
- Interosseous membrane
- Two intermuscular septa
- Direct attachment of the fascia crura to the periosteum of the tibia

Anterior compartment

- **Deep fibular nerve**
  - Dorsiflexion of foot
  - Inversion of the foot

Lateral compartment

- **Superficial fibular nerve**
  - Eversion of foot



Posterior compartment

- **Tibial nerve**
  - Plantarflexion of the foot
  - Inversion of the foot

Posterior leg

Summary of Posterior Compartment

MUSCLES
- gastrocnemius
- soleus
- plantaris
- tibialis posterior
- flexor digitorum longus
- flexor hallucis longus
- popliteus

MOTOR & SENSORY INNERVATION:
- tibial nerve

OPENINGS(CANAL)
- medial tarsal tunnel

BLOOD SUPPLY
- posterior tibial artery

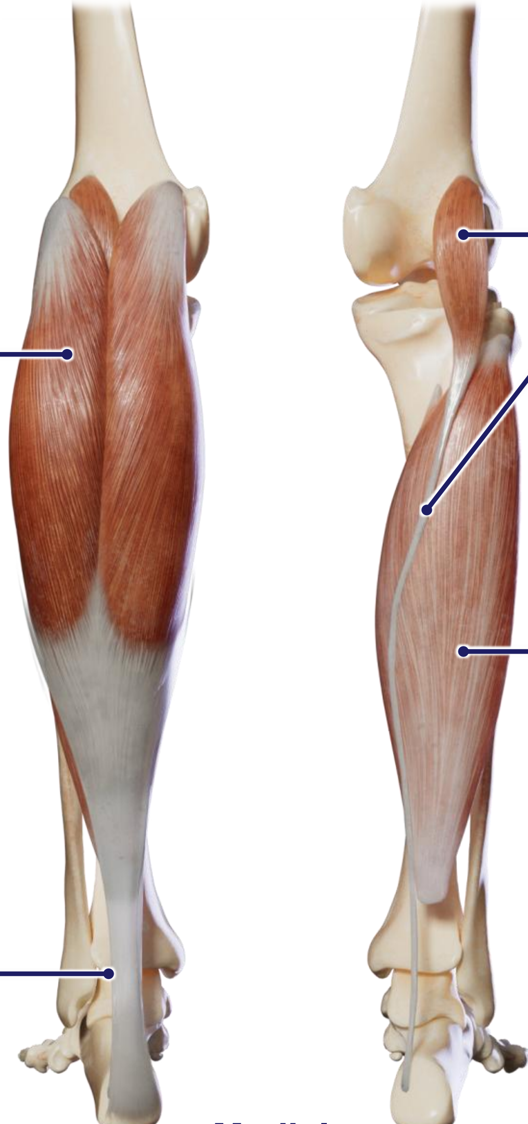
GENERAL FUNCTIONS
- plantarflexion of the ankle joint, tibialis posterior also inverts the ankle joint, flexion of digits

Posterior Leg Compartment Muscles - Superficial

Gastrocnemius
- Insertion: Achilles tendon
- **Tibial nerve**
- **Plantarflexes** ankle join/foot and flexes the knee/leg

Plantaris
- Insertion: Achilles tendon
- **Tibial nerve**
- **Plantarflexes** ankle join/foot and flexes the knee/leg

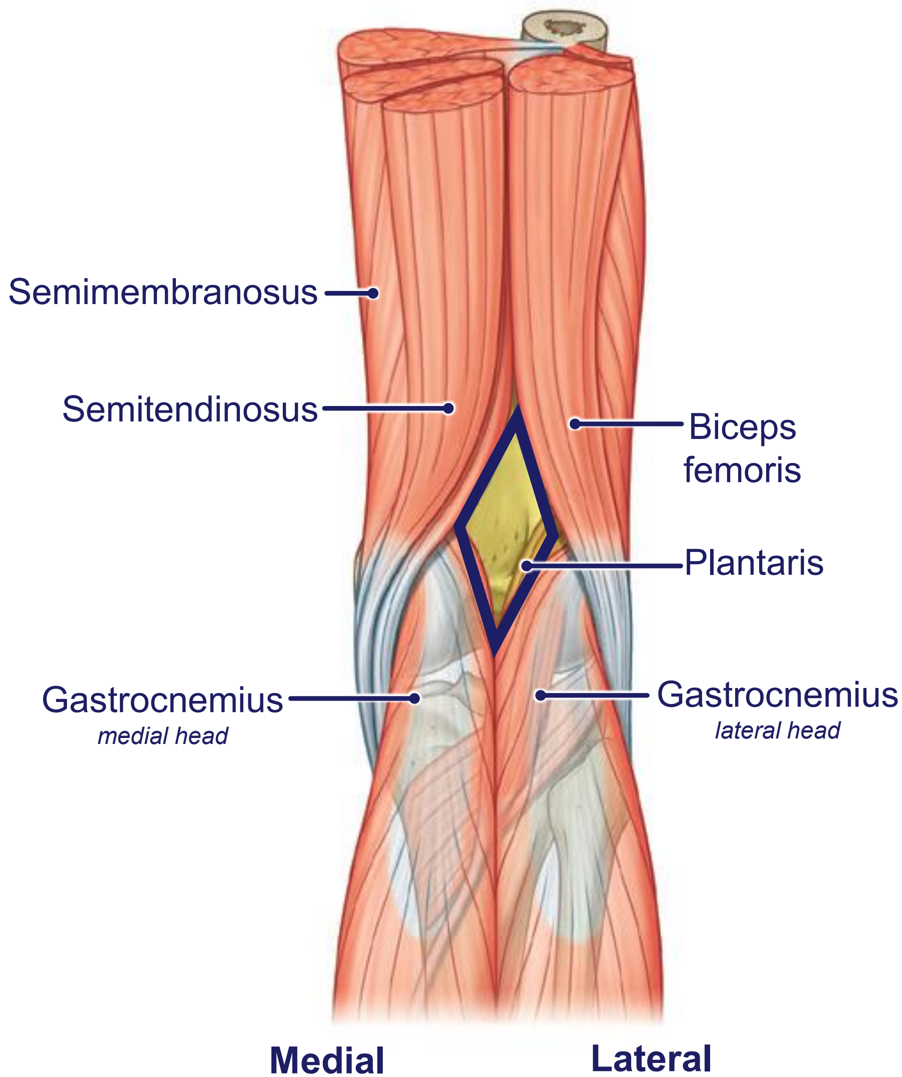
Soleus
- Insertion: Achilles tendon
- **Tibial nerve**
- **Plantarflexes** ankle join/foot



Popliteal Fossa

Diamond-shaped space behind the knee joint

- **Upper margins:**
  - Medially: semitendinosus and semimembranosus muscles
  - Laterally: biceps femoris muscle
- **Lower margins:**
  - Medially: medial head of the gastrocnemius muscle
  - Laterally: plantaris muscle and the lateral head of the gastrocnemius muscle
- **Floor:**
  - Capsule of the knee joint and adjacent surfaces of the femur and tibia,
  - More inferiorly, the popliteus muscle



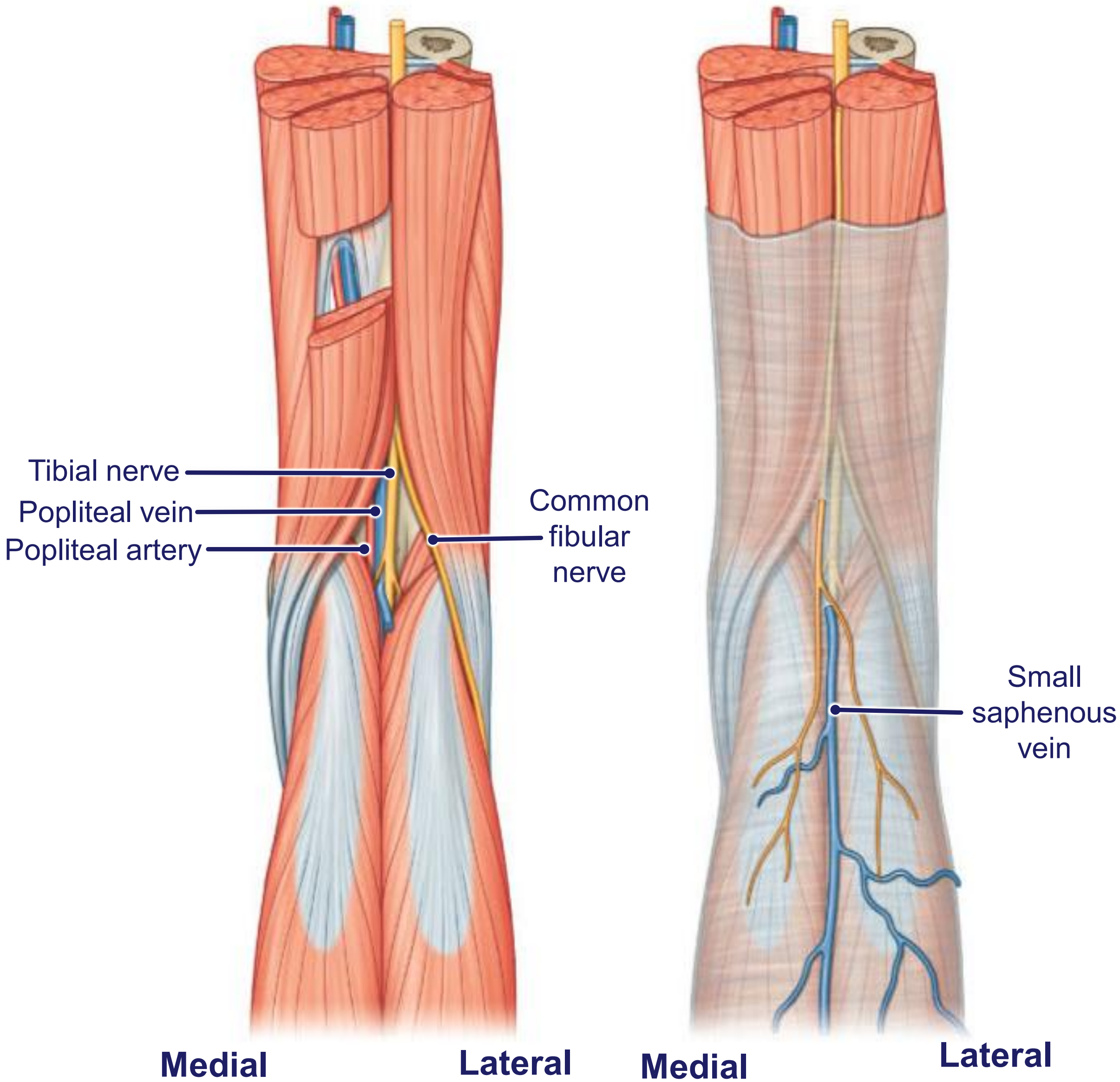
Popliteal Fossa

Arranged from superficial to deep:

- Tibial and common fibular nerves
- Popliteal vein
- Popliteal artery

Roof: covered by fascia and skin

- Small saphenous vein
- Posterior cutaneous nerve of the thigh

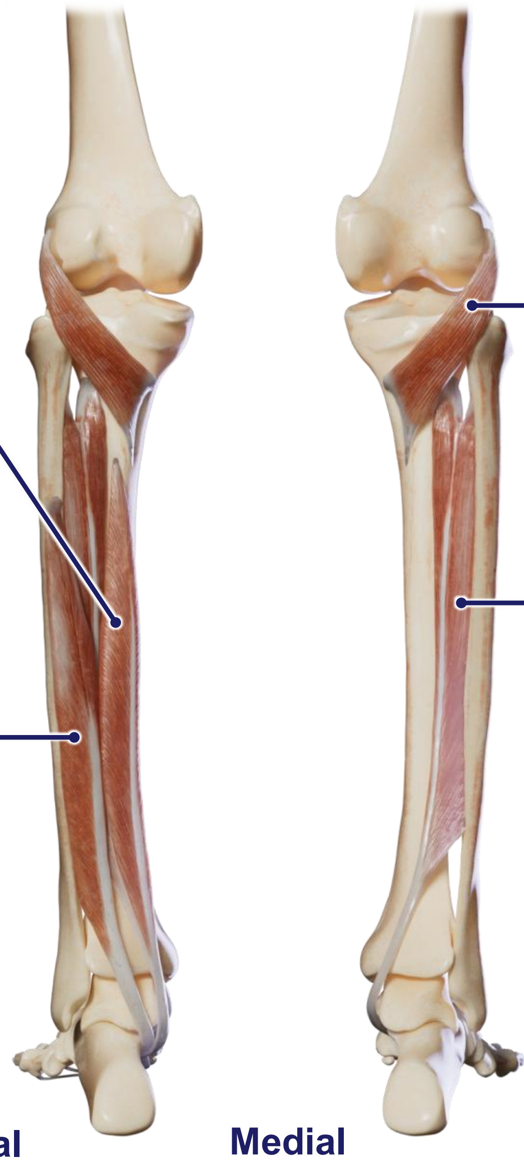


Posterior Leg Compartment Muscles – Deep

__NB:__ The tendons of **the tibialis posterior**, **flexor digitorum longus** and **flexor hallucis longus** enter the **plantar surface** of the foot through the **medial tarsal tunnel**

Flexor digitorum longus
- **Tibial nerve**
- **Flexes** lateral 4 toes

Flexor hallucis longus
- **Tibial nerve**
- **Flexes** the great toe



Popliteus
- **Tibial nerve**
- Unlocks the knee (laterally rotates femur on fixed tibia)

Tibialis posterior
- **Tibial nerve**
- **Plantarflexion** & **inversion** of ankle joint/foot
- Support arch of foot

Anterior leg

Summary of Anterior Compartment

MUSCLES
- tibialis anterior
- extensor digitorum longus
- extensor hallucis longus
- fibularis tertius

MOTOR & SENSORY INNERVATION:
- deep fibular nerve

BLOOD SUPPLY
- anterior tibial artery

GENERAL FUNCTIONS
- dorsiflexion of the ankle joint, tibialis anterior also inverts the ankle joint, extension of digits

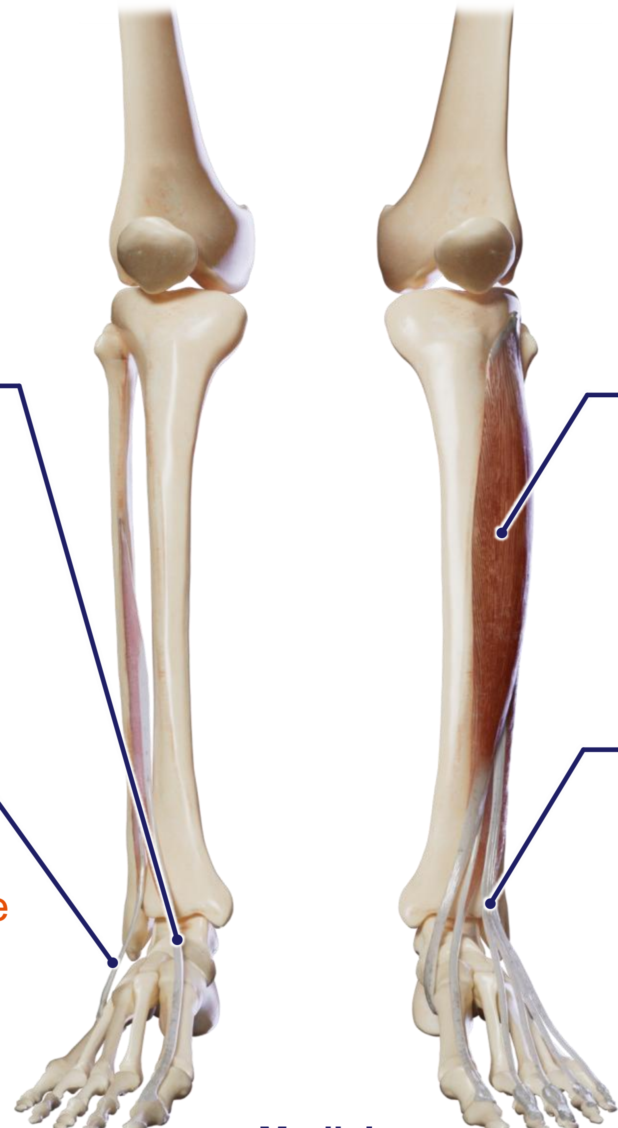
Anterior Compartment Muscles

Extensor hallucis longus
- **Deep fibular nerve**
- **Extension** of big toe
- **Dorsiflexion** of ankle joint/foot

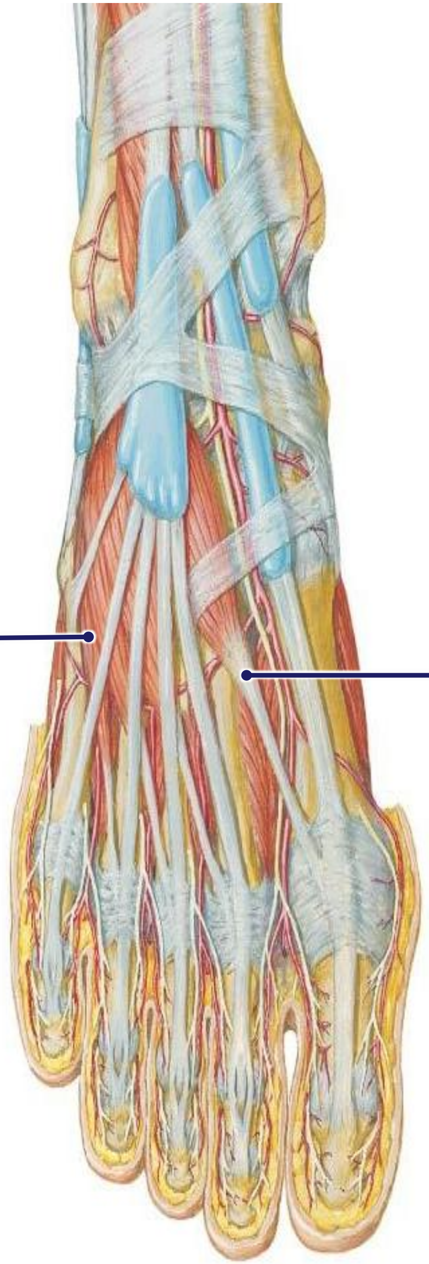
Fibularis tertius
- **Deep fibular nerve**
- **Dorsiflexion** of ankle joint/foot
- **Eversion** of ankle joint/foot

Tibialis anterior
- **Deep fibular nerve**
- **Dorsiflexion** of ankle joint/foot
- **Inversion** of ankle joint/foot
- Support arch of foot

Extensor digitorum longus
- **Deep fibular nerve**
- **Extension** of four lateral toes
- **Dorsiflexion** of ankle joint/ foot



Muscles of the Dorsum of Foot



Extensor digitorum brevis
- Deep fibular nerve
- **Extension** of digits II-IV

Extensor hallucis brevis
- Deep fibular nerve
- **Extension** of digit I

Lateral leg

Summary of Lateral Compartment

MUSCLES
- fibularis longus
- fibularis brevis

MOTOR & SENSORY INNERVATION:
- superficial fibular nerve

BLOOD SUPPLY
- fibular artery

GENERAL FUNCTIONS
- eversion of the ankle joint, supports the arch of the foot

Lateral Compartment Muscles

Fibularis longus

- **Insertion:** medial cuneiform and base of first metatarsal
- Superficial fibular nerve
- **Eversion** of ankle joint/foot
- Support arch of foot

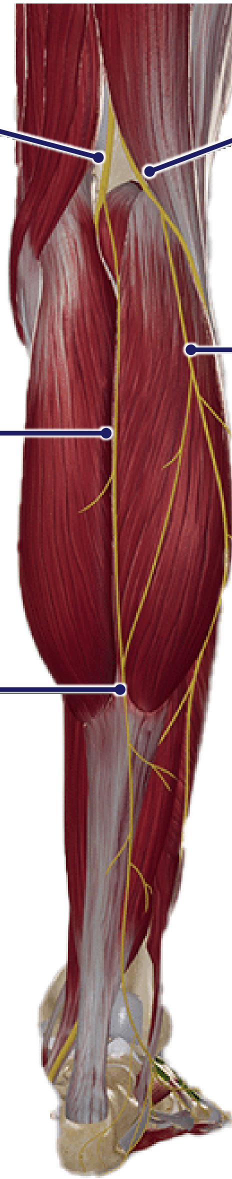
Lateral Compartment Muscles

Fibularis brevis

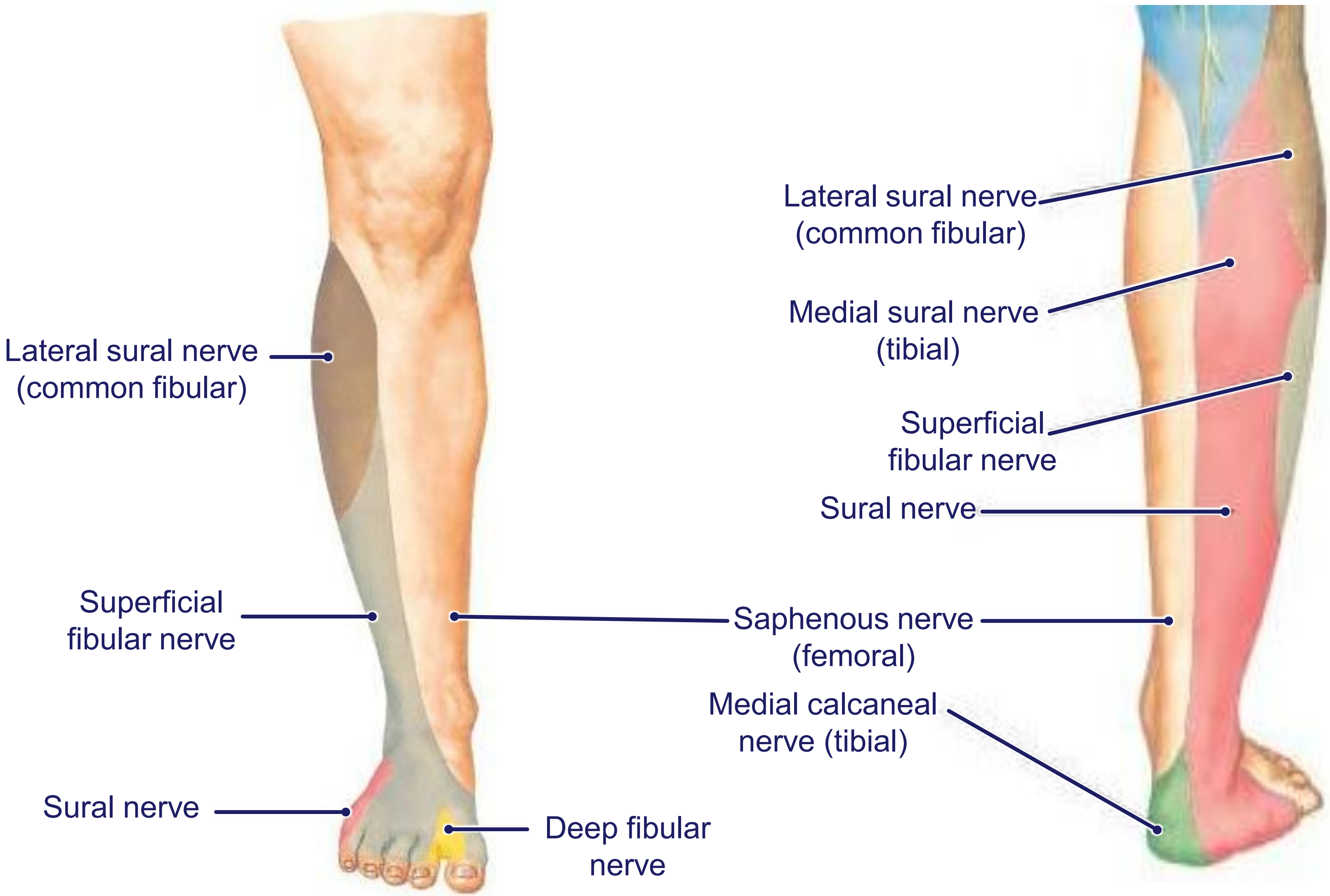
- Superficial fibular nerve
- **Eversion** of ankle joint/foot

Cutaneous innervation of the leg

Cutaneous Innervation of the Leg

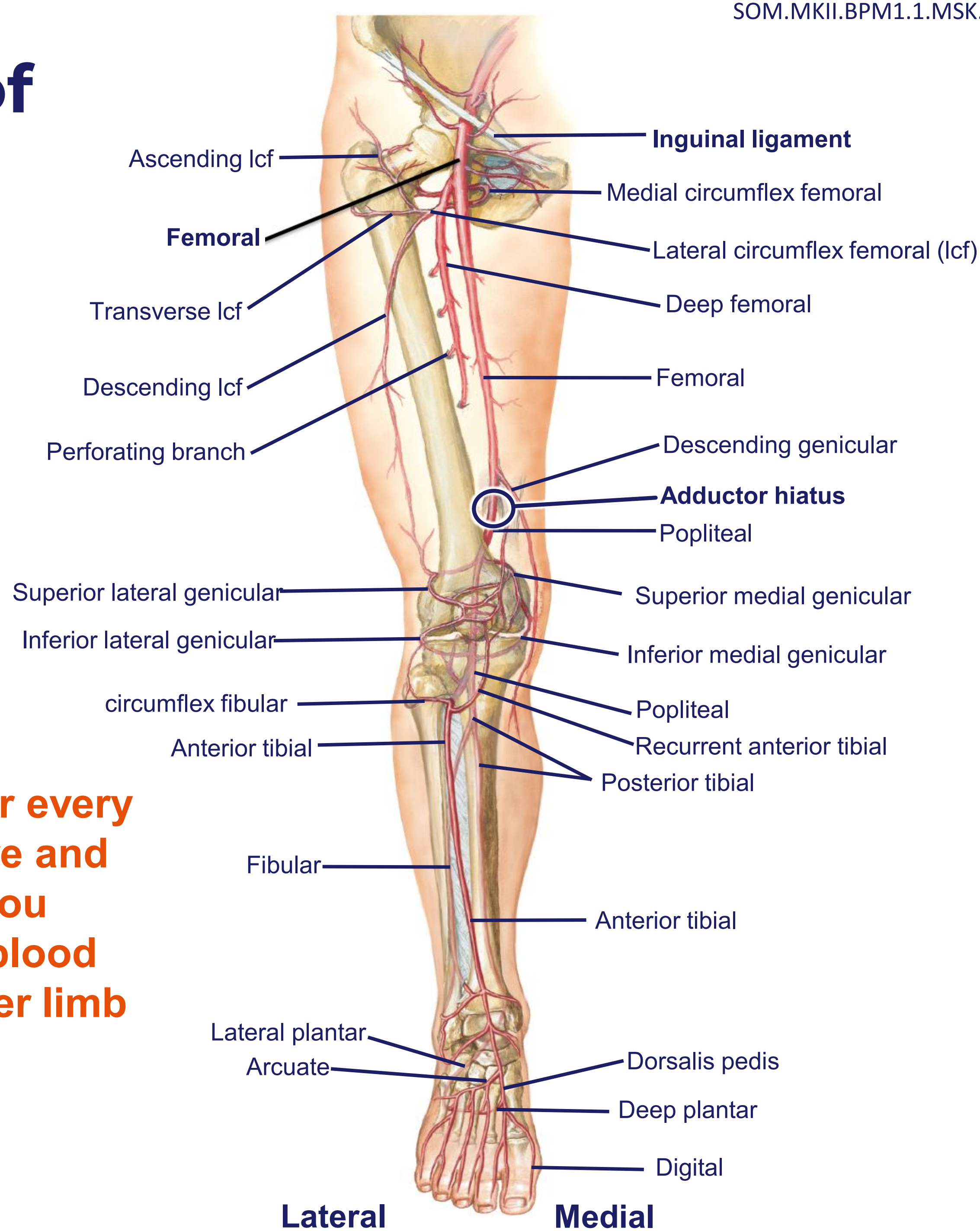


Cutaneous innervation of the Leg



Arterial blood supply of the leg

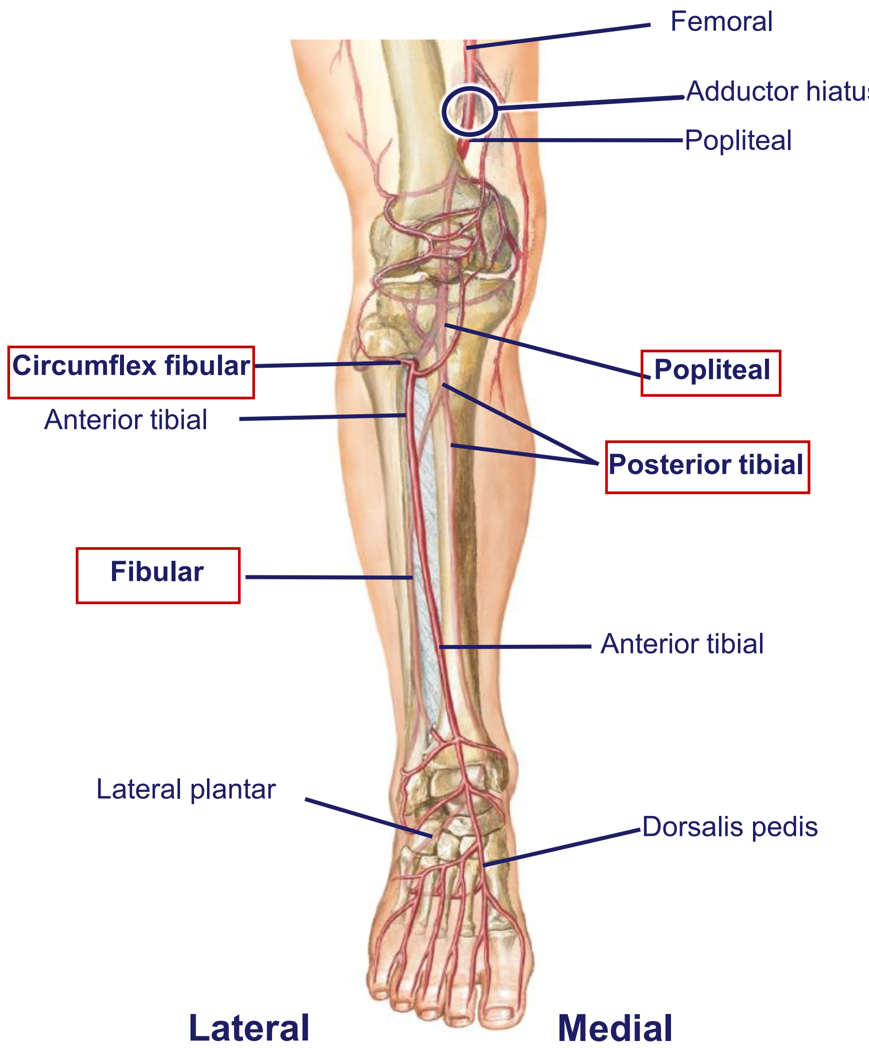
Summary of arterial supply of lower limb



Use this slide after every lower limb lecture and DLA to help you summarize the blood supply to the lower limb

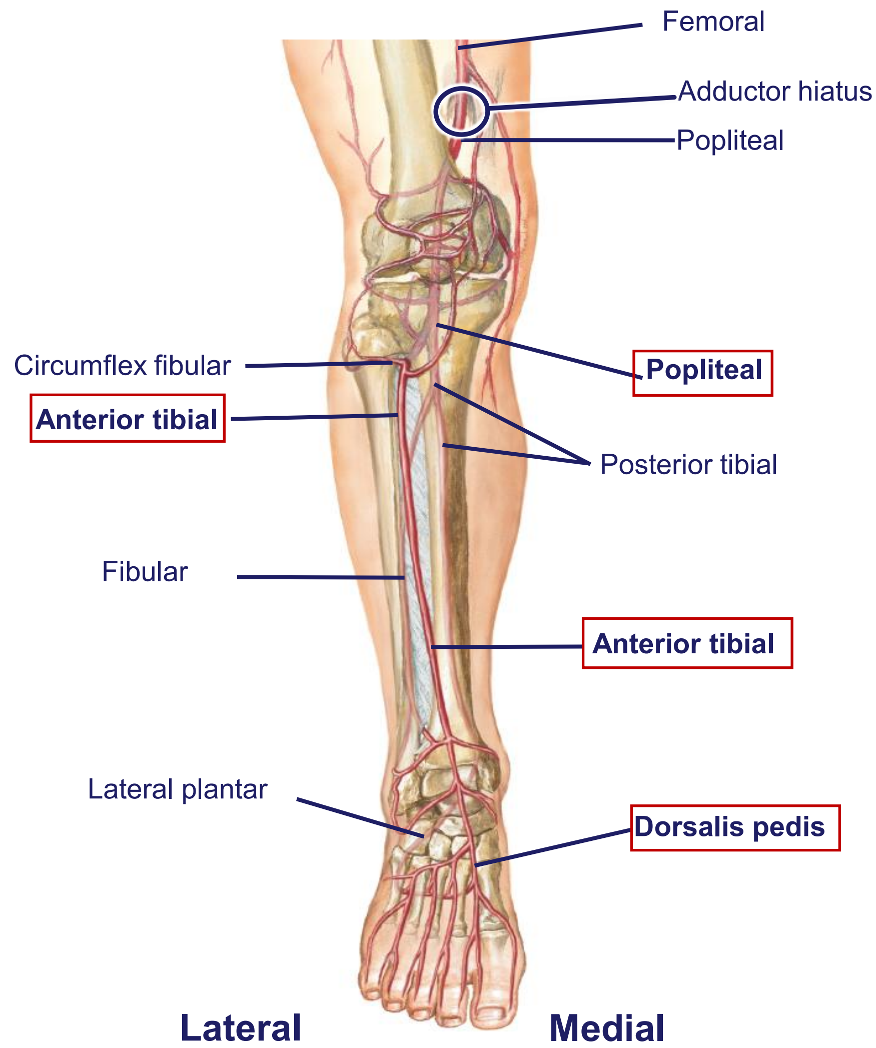
Arteries of the Leg

- **Posterior tibial artery:**
    - **Originates from the popliteal artery**
    - Enters and supplies the posterior compartment of the leg
    - Continues to the tarsal tunnel and splits into medial and lateral plantar arteries that supply the plantar surface of the foot
- **Gives the following branches:**
    - **Fibular:** perforating arteries supply the lateral compartment
    - **Circumflex fibular:** contributes to blood supply to knee joint

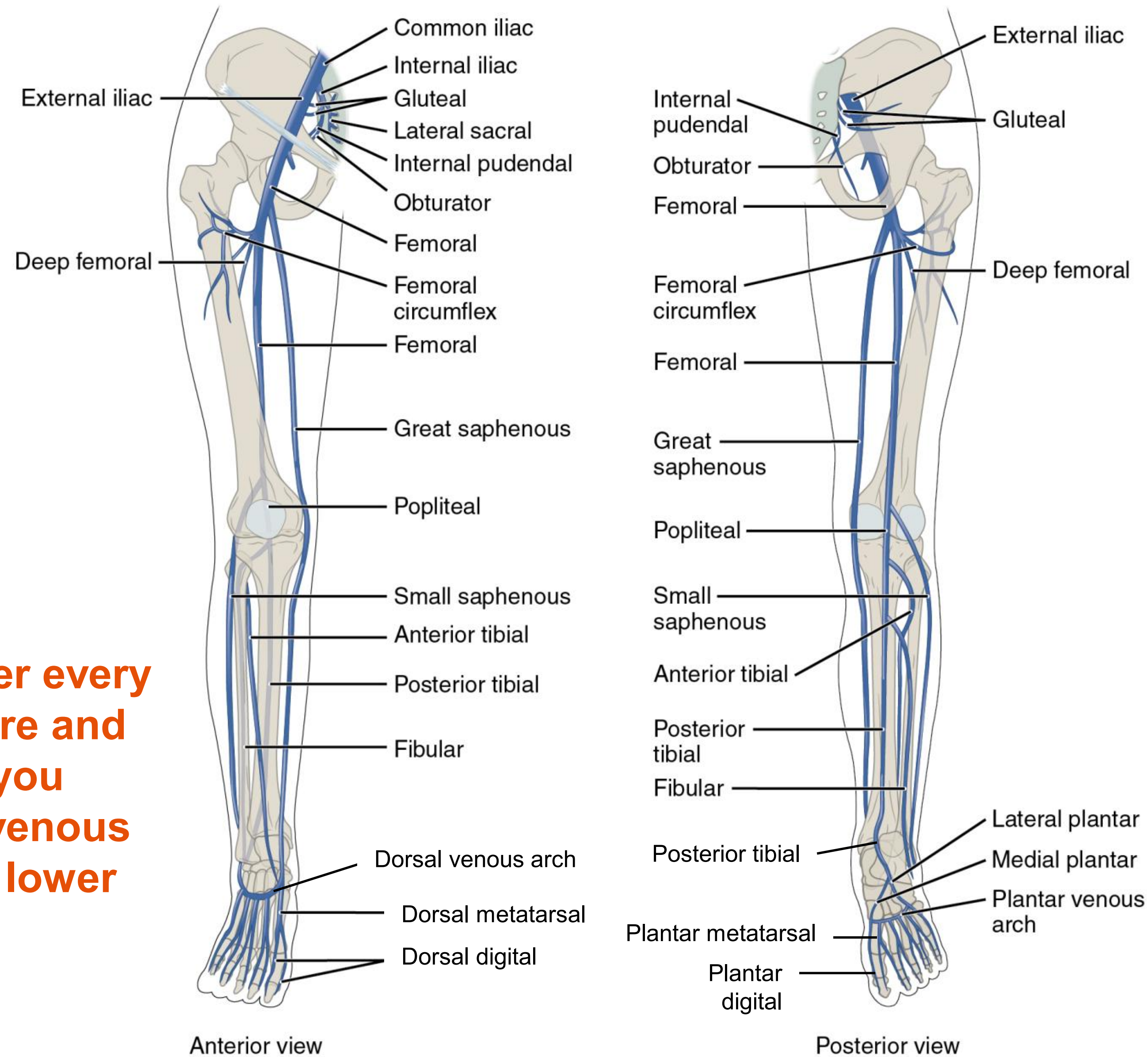


Arteries of the Leg

- **Anterior tibial artery:**
  - Originates from the popliteal artery
  - Enters and supplies the anterior compartment through an aperture in the interosseous membrane
  - Continues onto the dorsal aspect of the foot as the dorsalis pedis artery

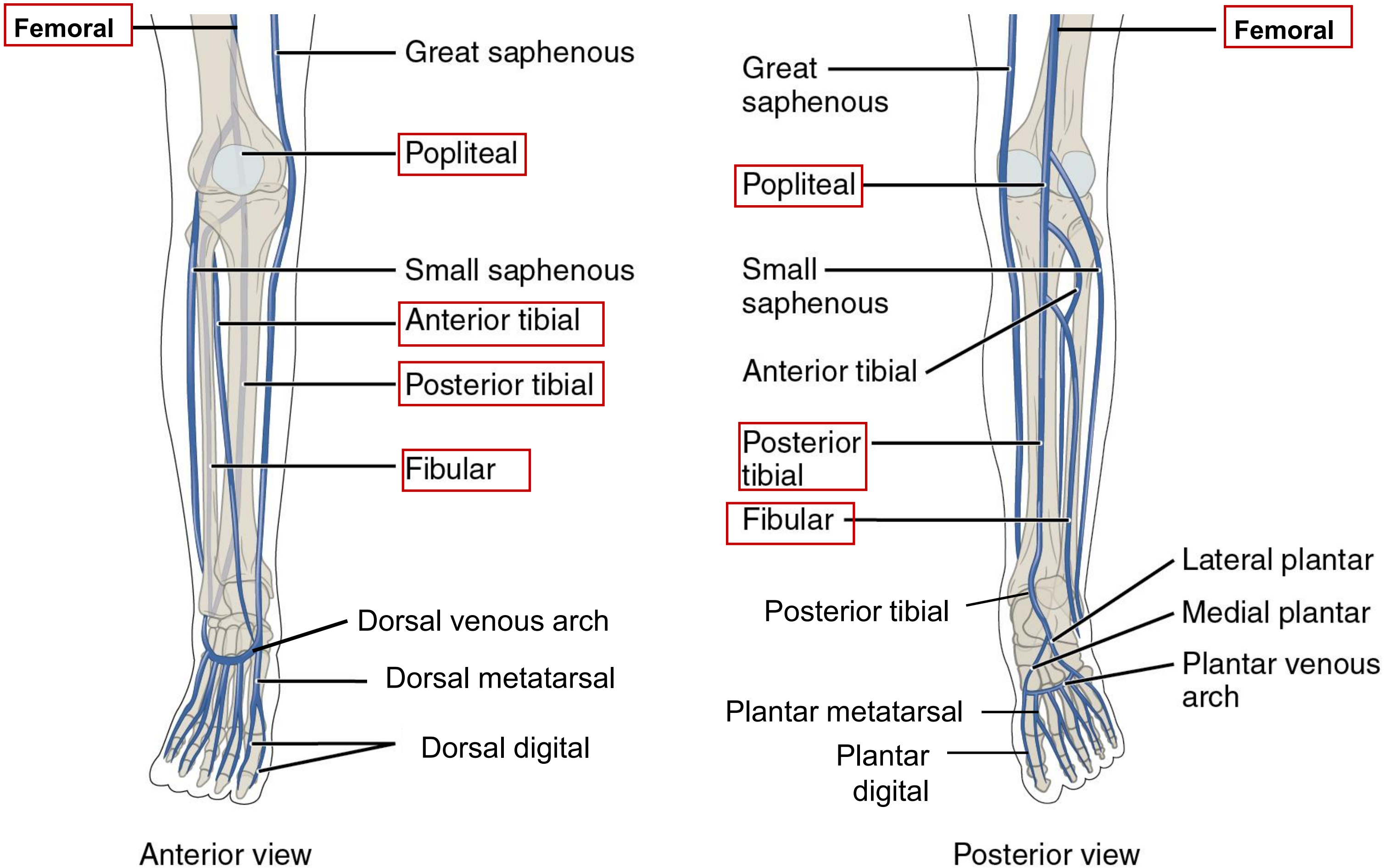


Summary of venous drainage of lower limb



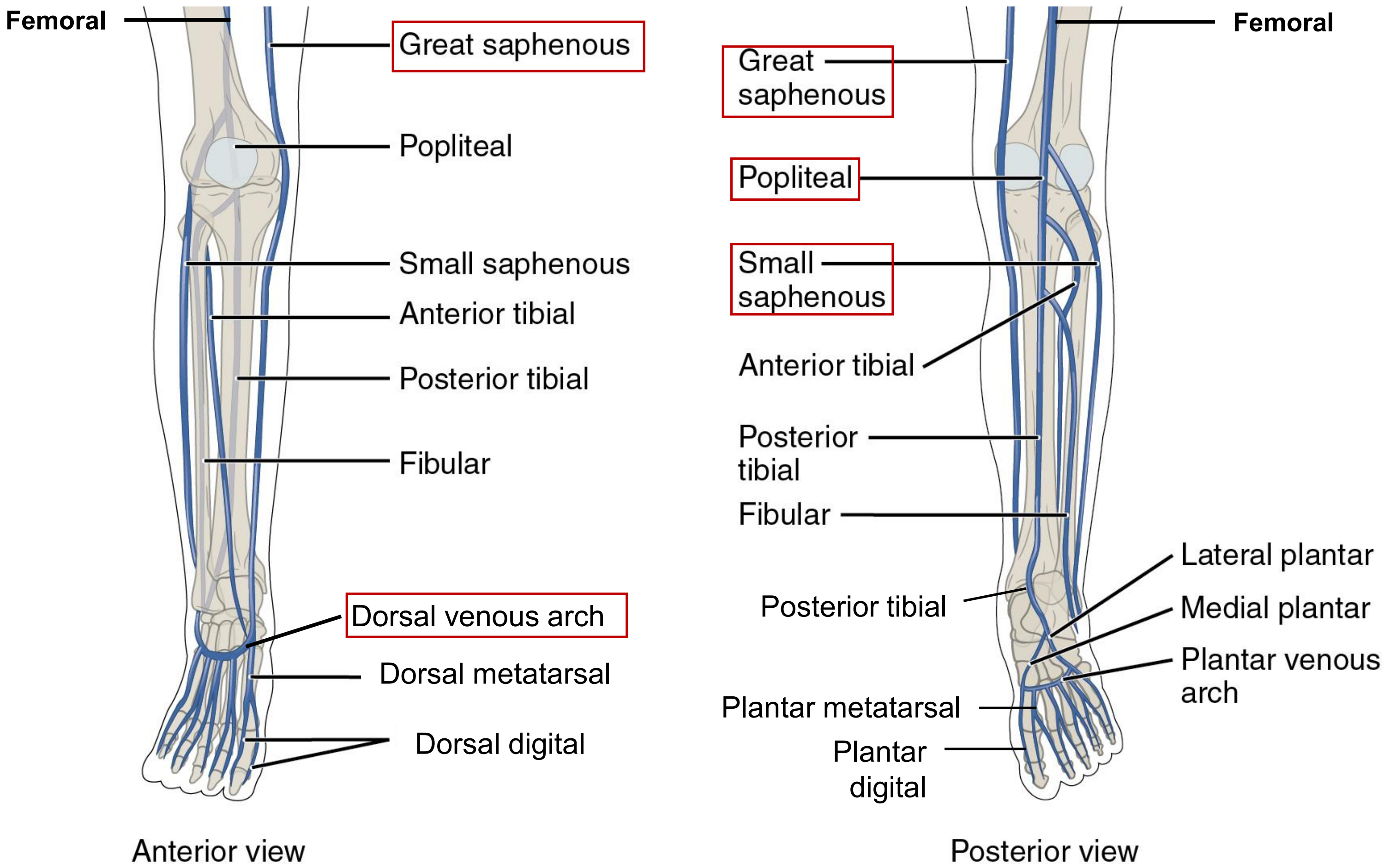
Use this slide after every lower limb lecture and DLA to help you summarize the venous drainage to the lower limb

Deep Veins of the Leg



- Located in compartments of leg
- **Fibular veins** drain into **posterior tibial vein**
- **Anterior and posterior tibial veins** join to become the **popliteal vein.**
- **Popliteal vein** becomes the **femoral vein** at the adductor hiatus

Superficial Veins of the Leg



- Located in the skin and superficial fascia of the leg.
- **Dorsal venous arch** becomes **small saphenous** on the lateral aspect of the foot. This vein runs posterior on the leg to drain into the **popliteal vein**. On the medial aspect of the foot, this arch becomes the **great saphenous vein** which runs on the medial side of the leg to the thigh. Drains into **femoral vein** at the saphenous opening.

Injury to the tibial and common fibular nerves

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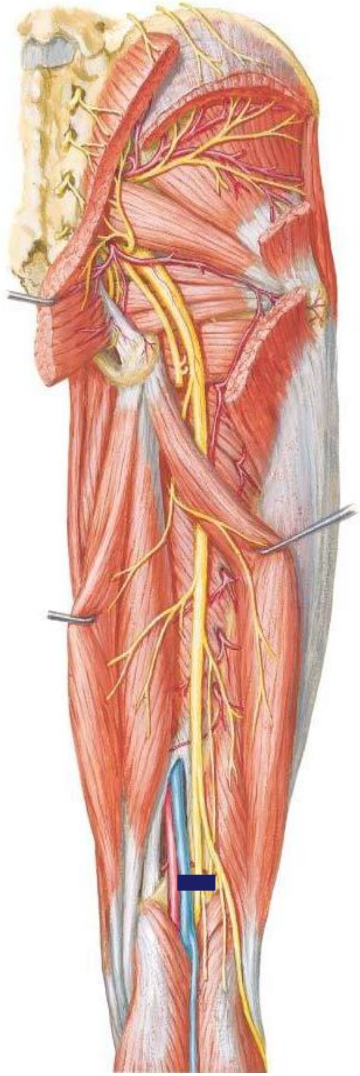
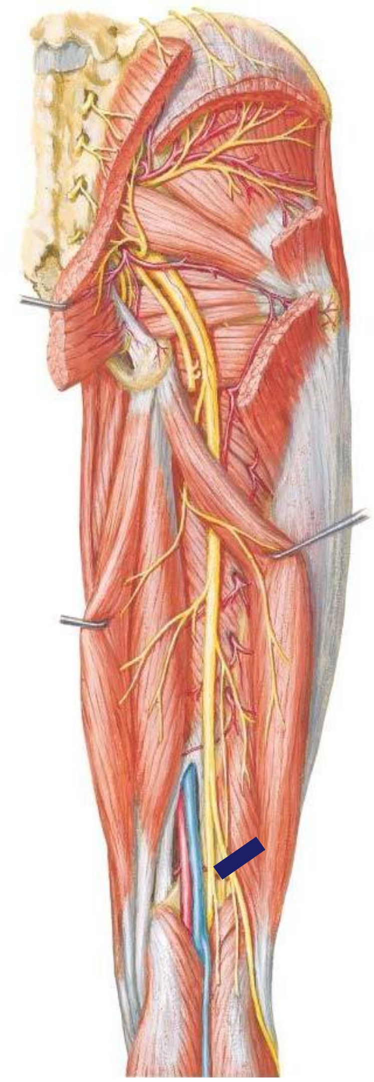
Tibial Nerve Injury

- Injury to the tibial nerve at its origin affects everything innervated by the tibial nerve.
  - **Muscles of the posterior compartment of the leg** and the **muscles in the plantar surface (sole) of the foot.**
  - Loss of plantarflexion the ankle joint.
  - Diminished inversion
  - All functions (except for extension) of the intrinsic joints of the feet will be lost
  - Loss of sensation on the posterolateral leg, heel and the sole of the foot



Common Fibular Nerve Injury

- Injury to the common fibular nerve at its origin affects everything innervated by the common fibular nerve
  - **Muscles of the lateral and anterior compartments of the leg** and the **muscles on the dorsum of the foot.**
  - Loss of dorsiflexion leading to foot drop
    - lower limb becomes 'too long' so to compensate, the patient will use a high stepping gait and/or swing out the leg
  - Loss of eversion
  - Diminished inversion
  - Loss of extension of the digits
  - Loss of sensation on the superolateral leg, posterolateral leg, antero-inferolateral leg and dorsum of foot



What would be the consequences of damage to the:

- *Deep fibular*
- *Superficial fibular*

Worksheet (for self-review)

Movement of the Ankle Joint	Muscles acting	Nerve innervating muscle
Dorsiflexion		
Plantarflexion		
Inversion		
Eversion		